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13. Bayesian Computation in Recurrent Neural Circuits

Rajesh P. N. Rao

https://doi.org/10.1162/08997660460733976 | pp. 1-38

A large number of human psychophysical results have been successfully explained in recent years using Bayesian models. However, the neural implementation of such models remains largely unclear. In this article, we show that a network architecture commonly used to model the cerebral cortex can implement Bayesian inference for an arbitrary hidden Markov model. We illustrate the approach using an orientation discrimination task and a visual motion detection task. In the case of orientation discrimination, we show that the model network can infer the posterior distribution over orientations and correctly estimate stimulus orientation in the presence of significant noise. In the case of motion detection, we show that the resulting model network exhibits direction selectivity and correctly computes the posterior probabilities over motion direction and position. When used to solve the well-known random dots motion discrimination task, the model generates responses that mimic the activities of evidence-accumulating neurons in cortical areas LIP and FEF. The framework we introduce posits a new interpretation of cortical activities in terms of log posterior probabilities of stimuli occurring in the natural world.

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14. How to Pretend That Correlated Variables Are Independent by Using Difference Observations

Christopher K. I. Williams

https://doi.org/10.1162/0899766052530884 | pp. 1-6

In many areas of data modeling, observations at different locations (e.g.,time frames or pixel locations) are augmented by differences of nea r by observations (e.g., delta features in speech recognition, Gabor jets in image analysis). These augmented observations are then often modeled as being independent. How can this make sense?We provide two interpretations,showing (1) that the likelihood of data generated from an auto regressive process can be computed in terms of "independent" augmented observations and (2) that the augmented observations can be given a coherent treatment in terms of the products of experts model (Hinton, 1999).

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15. Convergence of the IRWLS Procedure to the Support Vector Machine Solution

Fernando Pérez■Cruz, Carlos Bousoño■Calzón, Antonio Artés-Rodr■■guez

https://doi.org/10.1162/0899766052530875 | pp. 7-18

An iterative reweighted least squares (IRWLS) procedure recently proposed is shown to converge to the support vector machine solution. The convergence to a stationary point is ensured by modifying the original IRWLS procedure.

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16. Efficient Coding of Time-Relative Structure Using Spikes

Evan C. Smith, Michael S. Lewicki

https://doi.org/10.1162/0899766052530839 | pp. 19-45

Nonstationary acoustic features provide essential cues for many auditory tasks, including sound localization, auditory stream analysis, and speech recognition. These features can best be characterized relative to a precise point in time, such as the onset of a sound or the beginning of a harmonic periodicity. Extracting

these types of features is a difficult problem. Part of the difficulty is that with standard block-based signal analysis methods, the representation is sensitive to the arbitrary alignment of the blocks with respect to the signal. Convolutional techniques such as shift-invariant transformations can reduce this sensitivity, but these do not yield a code that is efficient, that is, one that forms a nonredundant representation of the underlying structure. Here, we develop a non-block-based method for signal representation that is both time relative and efficient. Signals are represented using a linear superposition of time-shiftable kernel functions, each with an associated magnitude and temporal position. Signal decomposition in this method is a non-linear process that consists of optimizing the kernel function scaling coefficients and temporal positions to form an efficient, shift-invariant representation. We demonstrate the properties of this representation for the purpose of characterizing structure in various types of nonstationary acoustic signals. The computational problem investigated here has direct relevance to the neural coding at the auditory nerve and the more general issue of how to encode complex, time-varying signals with a population of spiking neurons.

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17. Probability of Stimulus Detection in a Model Population of Rapidly Adapting Fibers

Burak Güçlü, Stanley J. Bolanowski

https://doi.org/10.1162/08997660460733985 | pp. 39-58

The goal of this study is to establish a link between somatosensory physiology and psychophysics at the probabilistic level. The model for a population of monkey rapidly adapting (RA) mechanoreceptive fibers by Güçlü and Bolanowski (2002) was used to study the probability of stimulus detection when a 40 Hz sinusoidal stimulation is applied with a constant contactor size (2 mm radius) on the terminal phalanx. In the model, the detection was assumed to be mediated by one or more active fibers. Two hypothetical receptive field organizations (uniformly random and gaussian) with varying average innervation densities were considered. At a given stimulus-contactor location, changing the stimulus amplitude generates sigmoid probability-of-detection curves for both receptive field organizations. The psychophysical results superimposed on these probability curves suggest that 5 to 10 active fibers may be required for detection. The effects of the contactor location on the probability of detection reflect the pattern of innervation in the model. However, the psychophysical data do not match with the predictions from the populations with uniform or gaussian distributed receptive field centers. This result may be due to some unknown mechanical factors along the terminal phalanx, or simply because a different receptive field organization is present. It has been reported that human observers can detect one single spike in an RA fiber. By considering the probability of stimulus detection across subjects and RA populations, this article proves that more than one active fiber is indeed required for detection.

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18. Bilinear Sparse Coding for Invariant Vision

David B. Grimes, Rajesh P. N. Rao

https://doi.org/10.1162/0899766052530893 | pp. 47-73

Recent algorithms for sparse coding and independent component analysis (ICA) have demonstrated how localized features can be learned from natural images. However, these approaches do not take image transformations into account. We describe an unsupervised algorithm for learning both localized features and their transformations directly from images using a sparse bilinear generative model. We show that from

an arbitrary set of natural images, the algorithm produces oriented basis filters that can simultaneously represent features in an image and their transformations. The learned generative model can be used to translate features to different locations, thereby reducing the need to learn the same feature at multiple locations, a limitation of previous approaches to sparse coding and ICA. Our results suggest that by explicitly modeling the interaction between local image features and their transformations, the sparse bilinear approach can provide a basis for achieving transformation-invariant vision.

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19. A Machine Learning Method for Extracting Symbolic Knowledge from Recurrent Neural Networks

Amir Taghizadeh Vahed, Christian W. Omlin

https://doi.org/10.1162/08997660460733994 | pp. 59-71

Neural networks do not readily provide an explanation of the knowledge stored in their weights as part of their information processing. Until recently, neural networks were considered to be black boxes, with the knowledge stored in their weights not readily accessible. Since then, research has resulted in a number of algorithms for extracting knowledge in symbolic form from trained neural networks. This article addresses the extraction of knowledge in symbolic form from recurrent neural networks trained to behave like deterministic finite-state automata (DFAs). To date, methods used to extract knowledge from such networks have relied on the hypothesis that networks' states tend to cluster and that clusters of network states correspond to DFA states. The computational complexity of such a cluster analysis has led to heuristics that either limit the number of clusters that may form during training or limit the exploration of the space of hidden recurrent state neurons. These limitations, while necessary, may lead to decreased fidelity, in which the extracted knowledge may not model the true behavior of a trained network, perhaps not even for the training set. The method proposed here uses a polynomial time, symbolic learning algorithm to infer DFAs solely from the observation of a trained network's input-output behavior. Thus, this method has the potential to increase the fidelity of the extracted knowledge.

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20. Orthogonality of Decision Boundaries in Complex-Valued Neural Networks

Tohru Nitta

https://doi.org/10.1162/08997660460734001 | pp. 73-97

This letter presents some results of an analysis on the decision boundaries of complex-valued neural networks whose weights, threshold values, input and output signals are all complex numbers. The main results may be summarized as follows. (1) A decision boundary of a single complex-valued neuron consists of two hypersurfaces that intersect orthogonally, and divides a decision region into four equal sections. The XOR problem and the detection of symmetry problem that cannot be solved with two-layered real-valued neural networks, can be solved by two-layered complex-valued neural networks with the orthogonal decision boundaries, which reveals a potent computational power of complex-valued neural nets. Furthermore, the fading equalization problem can be successfully solved by the two-layered complex-valued neural network with the highest generalization ability. (2) A decision boundary of a three-layered complex-valued neural network has the orthogonal property as a basic structure, and its two hypersurfaces approach orthogonality as all the net inputs to each hidden neuron grow. In particular, most of the decision boundaries in the

three-layered complex-valued neural network inetersect orthogonally when the network is trained using Complex-BP algorithm. As a result, the orthogonality of the decision boundaries improves its generalization ability. (3) The average of the learning speed of the Complex-BP is several times faster than that of the Real-BP. The standard deviation of the learning speed of the Complex-BP is smaller than that of the Real-BP. It seems that the complex-valued neural network and the related algorithm are natural for learning complex-valued patterns for the above reasons.

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21. A Probabilistic Framework for Region-Specific Remodeling of Dendrites in Three-Dimensional Neuronal Reconstructions

Rishikesh Narayanan, Anusha Narayan, Sumantra Chattarji

https://doi.org/10.1162/0899766052530811 | pp. 75-96

Dendritic arborization is an important determinant of single-neuron function as well as the circuitry among neurons. Dendritic trees undergo remodeling during development, aging, and many pathological conditions, with many of the morphological changes being confined to certain regions of the dendritic tree. In order to analyze the functional consequences of such region-specific dendritic remodeling, it is essential to develop techniques that can systematically manipulate three-dimensional reconstructions of neurons. Hence, in this study, we develop an algorithm that uses statistics from precise morphometric analyses to systematically remodel neuronal reconstructions. We use the distribution function of the ratio of two normal distributed random variables to specify the probabilities of remodeling along various regions of the dendritic arborization. We then use these probabilities to drive an iterative algorithm for manipulating the dendritic tree in a region-specific manner. As a test, we apply this framework to a well-characterized example of dendritic remodeling: stress-induced dendritic atrophy in hippocampal CA3 pyramidal cells. We show that our pruning algorithm is capable of eliciting atrophy that matches biological data from rodent models of chronic stress.

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22. Analysis of Cyclic Dynamics for Networks of Linear Threshold Neurons

Huajin Tang, Kay Chen Tan, Weinian Zhang

https://doi.org/10.1162/0899766052530820 | pp. 97-114

The network of neurons with linear threshold (LT) transfer functions is a prominent model to emulate the behavior of cortical neurons. The analysis of dynamic properties for LT networks has attracted growing interest, such as multistability and boundedness. However, not much is known about how the connection strength and external inputs are related to oscillatory behaviors. Periodic oscillation is an important characteristic that relates to nondivergence, which shows that the network is still bounded although unstable modes exist. By concentrating on a general parameterized two-cell network, theoretical results for geometrical properties and existence of periodic orbits are presented. Although it is restricted to two-dimensional systems, the analysis can provide a useful contribution to analyze cyclic dynamics of some specific LT networks of high dimension. As an application, it is extended to an important class of biologically motivated networks of large scale: the winner-take-all model using local excitation and global inhibition.

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23. On the Asymptotic Distribution of the Least-Squares Estimators in Unidentifiable Models

Taichi Hayasaka, Masashi Kitahara, Shiro Usui

https://doi.org/10.1162/08997660460734010 | pp. 99-114

In order to analyze the stochastic property of multilayered perceptrons or other learning machines, we deal with simpler models and derive the asymptotic distribution of the least-squares estimators of their parameters. In the case where a model is unidentified, we show different results from traditional linear models: the well-known property of asymptotic normality never holds for the estimates of redundant parameters.

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24. Asymptotic Properties of the Fisher Kernel

Koji Tsuda, Shotaro Akaho, Motoaki Kawanabe, Klaus■Robert Müller

https://doi.org/10.1162/08997660460734029 | pp. 115-137

This letter analyzes the Fisher kernel from a statistical point of view. The Fisher kernel is a particularly interesting method for constructing a model of the posterior probability that makes intelligent use of unlabeled data (i.e., of the underlying data density). It is important to analyze and ultimately understand the statistical properties of the Fisher kernel. To this end, we first establish sufficient conditions that the constructed posterior model is realizable (i.e., it contains the true distribution). Realizability immediately leads to consistency results. Subsequently, we focus on an asymptotic analysis of the generalization error, which elucidates the learning curves of the Fisher kernel and how unlabeled data contribute to learning. We also point out that the squared or log loss is theoretically preferable-because both yield consistent estimators-to other losses such as the exponential loss, when a linear classifier is used together with the Fisher kernel. Therefore, this letter underlines that the Fisher kernel should be viewed not as a heuristics but as a powerful statistical tool with well-controlled statistical properties.

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25. Nonlinear and Noisy Extension of Independent Component Analysis: Theory and Its Application to a Pitch Sensation Model

Shin■ichi Maeda, Wen■Jie Song, Shin Ishii

https://doi.org/10.1162/0899766052530866 | pp. 115-144

In this letter, we propose a noisy nonlinear version of independent component analysis (ICA). Assuming that the probability density function (p. d. f.) of sources is known, a learning rule is derived based on maximum likelihood estimation (MLE). Our model involves some algorithms of noisy linear ICA (e. g., Bermond & Cardoso, 1999) or noise-free nonlinear ICA (e. g., Lee, Koehler, & Orglmeister, 1997) as special cases. Especially when the nonlinear function is linear, the learning rule derived as a generalized expectation-maximization algorithm has a similar form to the noisy ICA algorithm previously presented by Douglas, Cichocki, and Amari (1998). Moreover, our learning rule becomes identical to the standard noise-free linear ICA algorithm in the noiseless limit, while existing MLE-based noisy ICA algorithms do not rigorously include the noise-free ICA. We trained our noisy nonlinear ICA by using acoustic signals such as speech and music. The model after learning successfully simulates virtual pitch phenomena, and the existence region of virtual pitch is qualitatively similar to that observed in a psychoacoustic experiment.

Although a linear transformation hypothesized in the central auditory system can account for the pitch sensation, our model suggests that the linear transformation can be acquired through learning from actual acoustic signals. Since our model includes a cepstrum analysis in a special case, it is expected to provide a useful feature extraction method that has often been given by the cepstrum analysis.

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26. Adaptive Hybrid Learning for Neural Networks

Rob Smithies, Sa■■d Salhi, N.M. Queen

https://doi.org/10.1162/08997660460734038 | pp. 139-157

A robust locally adaptive learning algorithm is developed via two enhancements of the Resilient Propagation (RPROP) method. Remaining drawbacks of the gradient-based approach are addressed by hybridization with gradient-independent Local Search. Finally, a global optimization method based on recursion of the hybrid is constructed, making use of tabu neighborhoods to accelerate the search for minima through diversification. Enhanced RPROP is shown to be faster and more accurate than the standard RPROP in solving classification tasks based on natural data sets taken from the UCI repository of machine learning databases. Furthermore, the use of Local Search is shown to improve Enhanced RPROP by solving the same classification tasks as part of the global optimization method.

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27. Online Ranking by Projecting

Koby Crammer, Yoram Singer

https://doi.org/10.1162/0899766052530848 | pp. 145-175

We discuss the problem of ranking instances. In our framework, each instance is associated with a rank or a rating, which is an integer in 1 to k. Our goal is to find a rank-prediction rule that assigns each instance a rank that is as close as possible to the instance's true rank. We discuss a group of closely related online algorithms, analyze their performance in the mistake-bound model, and prove their correctness. We describe two sets of experiments, with synthetic data and with the EachMovie data set for collaborative filtering. In the experiments we performed, our algorithms outperform online algorithms for regression and classification applied to ranking.

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28. Divergence Function, Duality, and Convex Analysis

Jun Zhang

https://doi.org/10.1162/08997660460734047 | pp. 159-195

From a smooth, strictly convex function phi: Rn --> R, a parametric family of divergence function Dphi(alpha) may be introduced: [equation: see text] for x, y epsilon int dom (Phi) subset Rn, and for alpha in R, with Dphi(+/-1) defined through taking the limit of alpha. Each member is shown to induce an alpha-independent Riemannian metric, as well as a pair of dual alpha-connections, which are generally nonflat, except for alpha = +/-1. In the latter case, Dphi(+/-1) reduces to the (nonparametric) Bregman divergence, which is representable using phi and its convex conjugate phi* and becomes the canonical divergence for dually flat spaces (Amari, 1982, 1985; Amari & Nagaoka, 2000). This formulation based on convex analysis naturally extends the informationgeometric interpretation of divergence functions (Eguchi, 1983) to allow the

distinction between two different kinds of duality: referential duality (alpha <--> -alpha) and representational duality (phi <--> phi*). When applied to (not necessarily normalized) probability densities, the concept of conjugated representations of densities is introduced, so that +/-alpha-connections defined on probability densities embody both referential and representational duality and are hence themselves bidual. When restricted to a finite-dimensional affine submanifold, the natural parameters of a certain representation of densities and the expectation parameters under its conjugate representation form biorthogonal coordinates. The alpha representation (indexed by beta now, beta epsilon [-1, 1]) is shown to be the only measure-invariant representation. The resulting two-parameter family of divergence functionals D(alpha,beta), (alpha, beta) epsilon [-1, 1] x [-1, 1] induces identical Fisher information but bidual alpha-connection pairs; it reduces in form to Amari's alpha-divergence family when alpha = +/-1 or when beta = 1, but to the family of Jensen difference (Rao, 1987) when beta = -1.

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29. On Learning Vector-Valued Functions

Charles A. Micchelli, Massimiliano Pontil

https://doi.org/10.1162/0899766052530802 | pp. 177-204

In this letter, we provide a study of learning in a Hilbert space of vectorvalued functions. We motivate the need for extending learning theory of scalar-valued functions by practical considerations and establish some basic results for learning vector-valued functions that should prove useful in applications. Specifically, we allow an output space Y to be a Hilbert space, and we consider a reproducing kernel Hilbert space of functions whose values lie in Y. In this setting, we derive the form of the minimal norm interpolant to a finite set of data and apply it to study some regularization functionals that are important in learning theory. We consider specific examples of such functionals corresponding to multiple-output regularization networks and support vector machines, for both regression and classification. Finally, we provide classes of operator-valued kernels of the dot product and translation-invariant type.

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30. Linear Response Algorithms for Approximate Inference in Graphical Models

Max Welling, Yee Whye Teh

https://doi.org/10.1162/08997660460734056 | pp. 197-221

Belief propagation (BP) on cyclic graphs is an efficient algorithm for computing approximate marginal probability distributions over single nodes and neighboring nodes in the graph. However, it does not prescribe a way to compute joint distributions over pairs of distant nodes in the graph. In this article, we propose two new algorithms for approximating these pairwise probabilities, based on the linear response theorem. The first is a propagation algorithm that is shown to converge if BP converges to a stable fixed point. The second algorithm is based on matrix inversion. Applying these ideas to gaussian random fields, we derive a propagation algorithm for computing the inverse of a matrix.

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31. RSPOP: Rough Set–Based Pseudo Outer-Product Fuzzy Rule Identification Algorithm

Kai Keng Ang, Chai Quek
https://doi.org/10.1162/0899766052530857 | pp. 205-243

System modeling with neuro-fuzzy systems involves two contradictory requirements: interpretability verses accuracy. The pseudo outer-product (POP) rule identification algorithm used in the family of pseudo outer-product-based fuzzy neural networks (POPFNN) suffered from an exponential increase in the number of identified fuzzy rules and computational complexity arising from high-dimensional data. This decreases the interpretability of the POPFNN in linguistic fuzzy modeling. This article proposes a novel rough set-based pseudo outer-product (RSPOP) algorithm that integrates the sound concept of knowledge reduction from rough set theory with the POP algorithm. The proposed algorithm not only performs feature selection through the reduction of attributes but also extends the reduction to rules without redundant attributes. As many possible reducts exist in a given rule set, an objective measure is developed for POPFNN to correctly identify the reducts that improve the inferred consequence. Experimental results are presented using published data sets and real-world application involving highway traffic flow prediction to evaluate the effectiveness of using the proposed algorithm to identify fuzzy rules in the POPFNN using compositional rule of inference and singleton fuzzifier (POPFNN-CRI(S)) architecture. Results showed that the proposed rough set-based pseudo outer-product algorithm reduces computational complexity, improves the interpretability of neuro-fuzzy systems by identifying significantly fewer fuzzy rules, and improves the accuracy of the POPFNN.

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32. Analyzing Neural Responses to Natural Signals: Maximally Informative Dimensions

Tatyana O. Sharpee, Nicole C. Rust, William Bialek
https://doi.org/10.1162/089976604322742010 | pp. 223-250

We propose a method that allows for a rigorous statistical analysis of neural responses to natural stimuli that are nongaussian and exhibit strong correlations. We have in mind a model in which neurons are selective for a small number of stimulus dimensions out of a high-dimensional stimulus space, but within this subspace the responses can be arbitrarily nonlinear. Existing analysis methods are based on correlation functions between stimuli and responses, but these methods are guaranteed to work only in the case of gaussian stimulus ensembles. As an alternative to correlation functions, we maximize the mutual information between the neural responses and projections of the stimulus onto low-dimensional subspaces. The procedure can be done iteratively by increasing the dimensionality of this subspace. Those dimensions that allow the recovery of all of the information between spikes and the full unprojected stimuli describe the relevant subspace. If the dimensionality of the relevant subspace indeed is small, it becomes feasible to map the neuron's input-output function even under fully natural stimulus conditions. These ideas are illustrated in simulations on model visual and auditory neurons responding to natural scenes and sounds, respectively.

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33. Rapid Temporal Modulation of Synchrony by Competition in Cortical Interneuron Networks

Paul Tiesinga, Terrence J. Sejnowski
https://doi.org/10.1162/089976604322742029 | pp. 251-275

The synchrony of neurons in extrastriate visual cortex is modulated by selective attention even when there are only small changes in firing rate (Fries, Reynolds, Rorie, & Desimone, 2001). We used Hodgkin-Huxley type models of cortical neurons to investigate the mechanism by which the degree of synchrony can be modulated independently of changes in firing rates. The synchrony of local networks of model cortical interneurons interacting through GABA(A) synapses was modulated on a fast timescale by selectively activating a fraction of the interneurons. The activated interneurons became rapidly synchronized and suppressed the activity of the other neurons in the network but only if the network was in a restricted range of balanced synaptic background activity. During stronger background activity, the network did not synchronize, and for weaker background activity, the network synchronized but did not return to an asynchronous state after synchronizing. The inhibitory output of the network blocked the activity of pyramidal neurons during asynchronous network activity, and during synchronous network activity, it enhanced the impact of the stimulus-related activity of pyramidal cells on receiving cortical areas (Salinas & Sejnowski, 2001). Synchrony by competition provides a mechanism for controlling synchrony with minor alterations in rate, which could be useful for information processing. Because traditional methods such as cross-correlation and the spike field coherence require several hundred milliseconds of recordings and cannot measure rapid changes in the degree of synchrony, we introduced a new method to detect rapid changes in the degree of coincidence and precision of spike timing.

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34. Dynamic Analyses of Information Encoding in Neural Ensembles

Riccardo Barbieri, Loren M. Frank, David P. Nguyen, Michael C. Quirk, Victor Solo, Matthew A. Wilson, Emery N. Brown

https://doi.org/10.1162/089976604322742038 | pp. 277-307

Neural spike train decoding algorithms and techniques to compute Shannon mutual information are important methods for analyzing how neural systems represent biological signals. Decoding algorithms are also one of several strategies being used to design controls for brain-machine interfaces. Developing optimal strategies to design decoding algorithms and compute mutual information are therefore important problems in computational neuroscience. We present a general recursive filter decoding algorithm based on a point process model of individual neuron spiking activity and a linear stochastic state-space model of the biological signal. We derive from the algorithm new instantaneous estimates of the entropy, entropy rate, and the mutual information between the signal and the ensemble spiking activity. We assess the accuracy of the algorithm by computing, along with the decoding error, the true coverage probability of the approximate 0.95 confidence regions for the individual signal estimates. We illustrate the new algorithm by reanalyzing the position and ensemble neural spiking activity of CA1 hippocampal neurons from two rats foraging in an open circular environment. We compare the performance of this algorithm with a linear filter constructed by the widely used reverse correlation method. The median decoding error for Animal 1 (2) during 10 minutes of open foraging was 5.9 (5.5) cm, the median entropy was 6.9 (7.0) bits, the median information was 9.4 (9.4) bits, and the true coverage probability for 0.95 confidence regions was 0.67 (0.75) using 34 (32) neurons. These findings improve significantly on our previous results and suggest an integrated approach to dynamically reading neural codes, measuring their properties, and quantifying the accuracy with which encoded information is extracted.

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35. A Network Model of Perceptual Suppression Induced by Transcranial Magnetic Stimulation

Yoichi Miyawaki, Masato Okada

https://doi.org/10.1162/089976604322742047 | pp. 309-331

We modeled the inhibitory effects of transcranial magnetic stimulation (TMS) on a neural population. TMS is a noninvasive technique, with high temporal resolution, that can stimulate the brain via a brief magnetic pulse from a coil placed on the scalp. Because of these advantages, TMS is extensively used as a powerful tool in experimental studies of motor, perception, and other functions in humans. However, the mechanisms by which TMS interferes with neural activities, especially in terms of theoretical aspects, are totally unknown. In this study, we focused on the temporal properties of TMS-induced perceptual suppression, and we computationally analyzed the response of a simple network model of a sensory feature detector system to a TMS-like perturbation. The perturbation caused the mean activity to transiently increase and then decrease for a long period, accompanied by a loss in the degree of activity localization. When the afferent input consisted of a dual phase, with a strong transient component and a weak sustained component, there was a critical latency period of the perturbation during which the network activity was completely suppressed and converged to the resting state. The range of the suppressive period increased with decreasing afferent input intensity and reached more than 10 times the time constant of the neuron. These results agree well with typical experimental data for occipital TMS and support the conclusion that dynamical interaction in a neural population plays an important role in TMS-induced perceptual suppression.

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36. Adaptive Two-Pass Median Filter Based on Support Vector Machines for Image Restoration

Tzu-Chao Lin, Pao-Ta Yu

https://doi.org/10.1162/089976604322742056 | pp. 332-353

In this letter, a novel adaptive filter, the adaptive two-pass median (ATM) filter based on support vector machines (SVMs), is proposed to preserve more image details while effectively suppressing impulse noise for image restoration. The proposed filter is composed of a noise decision maker and two-pass median filters. Our new approach basically uses an SVM impulse detector to judge whether the input pixel is noise. If a pixel is detected as a corrupted pixel, the noise-free reduction median filter will be triggered to replace it. Otherwise, it remains unchanged. Then, to improve the quality of the restored image, a decision impulse filter is put to work in the second-pass filtering procedure. As for the noise suppressing both fixed-valued and random-valued impulses without degrading the quality of the fine details, the results of our extensive experiments demonstrate that the proposed filter outperforms earlier median-based filters in the literature. Our new filter also provides excellent robustness at various percentages of impulse noise.

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37. Adaptive Two-Pass Median Filter Based on Support Vector Machines for Image Restoration

Tzu-Chao Lin, Pao-Ta Yu

https://doi.org/10.1162/neco.2004.16.2.333 | pp. 333-354

In this letter, a novel adaptive filter, the adaptive two-pass median (ATM) filter based on support vector machines (SVMs), is proposed to preserve more image details while effectively suppressing impulse noise for image restoration. The proposed filter is composed of a noise decision maker and two-pass median filters. Our new approach basically uses an SVM impulse detector to judge whether the input pixel is noise. If a pixel is detected as a corrupted pixel, the noise-free reduction median filter will be triggered to replace it. Otherwise, it remains unchanged. Then, to improve the quality of the restored image, a decision impulse filter is put to work in the second-pass filtering procedure. As for the noise suppressing both fixed-valued and random-valued impulses without degrading the quality of the fine details, the results of our extensive experiments demonstrate that the proposed filter outperforms earlier median-based filters in the literature. Our new filter also provides excellent robustness at various percentages of impulse noise.

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38. Improving Generalization Performance of Natural Gradient Learning Using Optimized Regularization by NIC

Hyeyoung Park, Noboru Murata, ■■■-■■■ ■■■■■

https://doi.org/10.1162/089976604322742065 | pp. 355-382

Natural gradient learning is known to be efficient in escaping plateau, which is a main cause of the slow learning speed of neural networks. The adaptive natural gradient learning method for practical implementation also has been developed, and its advantage in real-world problems has been confirmed. In this letter, we deal with the generalization performances of the natural gradient method. Since natural gradient learning makes parameters fit to training data quickly,the overfitting phenomenon may easily occur, which results in poor generalization performance. To solve the problem, we introduce the regularization term in natural gradient learning and propose an efficient optimizing method for the scale of regularization by using a generalized Akaike information criterion (network information criterion). We discuss the properties of the optimized regularization strength by NIC through theoretical analysis as well as computer simulations. We confirm the computational efficiency and generalization performance of the proposed method in real-world applications through computational experiments on benchmark problems.

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39. One-Bit-Matching Conjecture for Independent Component Analysis

Zhiyong Liu, Kai-Chun Chiu, Lei Xu

https://doi.org/10.1162/089976604322742074 | pp. 383-399

The one-bit-matching conjecture for independent component analysis (ICA) could be understood from different perspectives but is basically stated as "all the sources can be separated as long as there is a one-to-one same-sign-correspondence between the kurtosis signs of all source probability density functions (pdf's) and the kurtosis signs of all model pdf's" (Xu, Cheung, & Amari, 1998a). This conjecture has been widely believed in the ICA community and implicitly supported by many ICA studies, such as the Extended Infomax (Lee, Girolami, & Sejnowski, 1999) and the soft switching algorithm (Welling & Weber, 2001). However, there is no mathematical proof to confirm the conjecture theoretically. In this article, only skewness and kurtosis are considered, and such a mathematical proof is given under the assumption that the skewness of the model densities vanishes. Moreover, empirical experiments are demonstrated on the robustness of the conjecture as the vanishing skewness assumption breaks. As a by-product, we also show that the kurtosis maximization criterion (Moreau & Macchi, 1996) is actually a special case of the minimum

mutual information criterion for ICA.

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40. Jacobian Conditioning Analysis for Model Validation

Isabelle Rivals, L. Personnaz

https://doi.org/10.1162/089976604322742083 | pp. 401-418

Our aim is to stress the importance of Jacobian matrix conditioning for model validation. We also comment on Monari and Dreyfus (2002), where, following Rivals and Personnaz (2000), it is proposed to discard neural candidates that are likely to overfit and/or for which quantities of interest such as confidence intervals cannot be computed accurately. In Rivals and Personnaz (2000), we argued that such models are to be discarded on the basis of the condition number of their Jacobian matrix. But Monari and Dreyfus (2002) suggest making the decision on the basis of the computed values of the leverages, the diagonal elements of the projection matrix on the range of the Jacobian, or "hat" matrix: they propose to discard a model if computed leverages are outside some theoretical bounds, pretending that it is the symptom of the Jacobian rank deficiency. We question this proposition because, theoretically, the hat matrix is defined whatever the rank of the Jacobian and because, in practice, the computed leverages of very ill-conditioned networks may respect their theoretical bounds while confidence intervals cannot be estimated accurately enough, two facts that have escaped Monari and Dreyfus's attention. Wealso recall the most accurate way to estimate the leverages and the properties of these estimations. Finally, we make an additional comment concerning the performance estimation in Monari and Dreyfus (2002).

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41. Reply to the Comments on “Local Overfitting Control via Leverages” in “Jacobian Conditioning Analysis for Model Validation” by I. Rivals and L. Personnaz

Yacine Oussar, Gaétan Monari, Gérard Dreyfus

https://doi.org/10.1162/089976604322742092 | pp. 419-443

"Jacobian Conditioning Analysis for Model Validation" by Rivals and Personnaz in this issue is a comment on Monari and Dreyfus (2002). In this reply, we disprove their claims. We point to flawed reasoning in the theoretical comments and to errors and inconsistencies in the numerical examples. Our replies are substantiated by seven counterexamples, inspired by actual data, which show that the comments on the accuracy of the computation of the leverages are unsupported and that following the approach they advocate leads to discarding valid models or validating overfitted models.

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42. Geometrical Computations Explain Projection Patterns of Long-Range Horizontal Connections in Visual Cortex

Ohad Ben■Shahar, Steven W. Zucker

https://doi.org/10.1162/089976604772744866 | pp. 445-476

Neurons in primary visual cortex respond selectively to oriented stimuli such as edges and lines. The long-range horizontal connections between them are thought to facilitate contour integration. While many physiological and psychophysical findings suggest that collinear or association field models of good

continuation dictate particular projection patterns of horizontal connections to guide this integration process, significant evidence of interactions inconsistent with these hypotheses is accumulating. We first show that natural random variations around the collinear and association field models cannot account for these inconsistencies, a fact that motivates the search for more principled explanations. We then develop a model of long-range projection fields that formalizes good continuation based on differential geometry. The analysis implicates curvature(s) in a fundamental way, and the resulting model explains both consistent data and apparent outliers. It quantitatively predicts the (typically ignored) spread in projection distribution, its nonmonotonic variance, and the differences found among individual neurons. Surprisingly, and for the first time, this model also indicates that texture (and shading) continuation can serve as alternative and complementary functional explanations to contour integration. Because current anatomical data support both (curve and texture) integration models equally and because both are important computationally, new testable predictions are derived to allow their differentiation and identification.

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43. Mean Instantaneous Firing Frequency Is Always Higher Than the Firing Rate

Petr Lánský, Roger Rodriguez, Laura Sacerdote

https://doi.org/10.1162/089976604772744875 | pp. 477-489

Frequency coding is considered one of the most common coding strategies employed by neural systems. This fact leads, in experiments as well as theoretical studies, to construction of so-called transfer functions, where the output firing frequency is plotted against the input intensity. The term "firing frequency" can be understood differently in different contexts. Basically, it means that the number of spikes over an interval of preselected length is counted and then divided by the length of the interval, but due to the obvious limitations, the length of observation cannot be arbitrarily long. Then firing frequency is defined as reciprocal to the mean interspike interval. In parallel, an instantaneous firing frequency can be defined as reciprocal to the length of current interspike interval, and by taking a mean of these, the definition can be extended to introduce the mean instantaneous firing frequency. All of these definitions of firing frequency are compared in an effort to contribute to a better understanding of the input-output properties of a neuron.

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44. Kalman Filter Control Embedded into the Reinforcement Learning Framework

István Szita, András Lörincz

https://doi.org/10.1162/089976604772744884 | pp. 491-499

There is a growing interest in using Kalman filter models in brain modeling. The question arises whether Kalman filter models can be used on-line not only for estimation but for control. The usual method of optimal control of Kalman filter makes use of off-line backward recursion, which is not satisfactory for this purpose. Here, it is shown that a slight modification of the linear-quadratic-gaussian Kalman filter model allows the on-line estimation of optimal control by using reinforcement learning and overcomes this difficulty. Moreover, the emerging learning rule for value estimation exhibits a Hebbian form, which is weighted by the error of the value estimation.

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45. Rapid Processing and Unsupervised Learning in a Model of the Cortical Macrocolumn

Jörg Lücke, Christoph von der Malsburg

https://doi.org/10.1162/089976604772744893 | pp. 501-533

We study a model of the cortical macrocolumn consisting of a collection of inhibitorily coupled minicolumns. The proposed system overcomes several severe deficits of systems based on single neurons as cerebral functional units, notably limited robustness to damage and unrealistically large computation time. Motivated by neuroanatomical and neurophysiological findings, the utilized dynamics is based on a simple model of a spiking neuron with refractory period, fixed random excitatory interconnection within minicolumns, and instantaneous inhibition within one macrocolumn. A stability analysis of the system's dynamical equations shows that minicolumns can act as monolithic functional units for purposes of critical, fast decisions and learning. Oscillating inhibition (in the gamma frequency range) leads to a phase-coupled population rate code and high sensitivity to small imbalances in minicolumn inputs. Minicolumns are shown to be able to organize their collective inputs without supervision by Hebbian plasticity into selective receptive field shapes, thereby becoming classifiers for input patterns. Using the bars test, we critically compare our system's performance with that of others and demonstrate its ability for distributed neural coding.

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46. Temporally Asymmetric Learning Supports Sequence Processing in Multi-Winner Self-Organizing Maps

Reiner Schulz, James A. Reggia

https://doi.org/10.1162/089976604772744901 | pp. 535-561

We examine the extent to which modified Kohonen self-organizing maps (SOMs) can learn unique representations of temporal sequences while still supporting map formation. Two biologically inspired extensions are made to traditional SOMs: selection of multiple simultaneous rather than single "winners" and the use of local intramap connections that are trained according to a temporally asymmetric Hebbian learning rule. The extended SOM is then trained with variable-length temporal sequences that are composed of phoneme feature vectors, with each sequence corresponding to the phonetic transcription of a noun. The model transforms each input sequence into a spatial representation (final activation pattern on the map). Training improves this transformation by, for example, increasing the uniqueness of the spatial representations of distinct sequences, while still retaining map formation based on input patterns. The closeness of the spatial representations of two sequences is found to correlate significantly with the sequences' similarity. The extended model presented here raises the possibility that SOMs may ultimately prove useful as visualization tools for temporal sequences and as preprocessors for sequence pattern recognition systems.

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47. Neuronal Bases of Perceptual Learning Revealed by a Synaptic Balance Scheme

Osamu Hoshino

https://doi.org/10.1162/089976604772744910 | pp. 563-594

Our ability to perceive external sensory stimuli improves as we experience the same stimulus repeatedly. This perceptual enhancement, called perceptual learning, has been demonstrated for various sensory systems, such as vision, audition, and somatosensation. I investigated the contribution of lateral excitatory and inhibitory synaptic balance to perceptual learning. I constructed a simple associative neural network model in which sensory features were expressed by the activities of specific cell assemblies. Each neuron is sensitive to a specific sensory feature, and the neurons belonging to the same cell assembly are sensitive to the same feature. During perceptual learning processes, the network was presented repeatedly with a stimulus that was composed of a sensory feature and noise, and the lateral excitatory and inhibitory synaptic connection strengths between neurons were modified according to a pulse-timing-based Hebbian rule. Perceptual learning enhanced the cognitive performance of the network, increasing the signal-to-noise ratio of neuronal activity. I suggest here that the alteration of the synaptic balance may be essential for perceptual learning, especially when the brain tries to adopt the most suitable strategy--signal enhancement, noise reduction, or both--for a given perceptual task.

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48. How the Shape of Pre- and Postsynaptic Signals Can Influence STDP: A Biophysical Model

Aušra Saudargien■, Bernd Porr, Florentin Wörgötter

https://doi.org/10.1162/089976604772744929 | pp. 595-625

Spike-timing-dependent plasticity (STDP) is described by long-term potentiation (LTP), when a presynaptic event precedes a postsynaptic event, and by long-term depression (LTD), when the temporal order is reversed. In this article, we present a biophysical model of STDP based on a differential Hebbian learning rule (ISO learning). This rule correlates presynaptically the NMDA channel conductance with the derivative of the membrane potential at the synapse as the postsynaptic signal. The model is able to reproduce the generic STDP weight change characteristic. We find that (1) The actual shape of the weight change curve strongly depends on the NMDA channel characteristics and on the shape of the membrane potential at the synapse. (2) The typical antisymmetrical STDP curve (LTD and LTP) can become similar to a standard Hebbian characteristic (LTP only) without having to change the learning rule. This occurs if the membrane depolarization has a shallow onset and is long lasting. (3) It is known that the membrane potential varies along the dendrite as a result of the active or passive backpropagation of somatic spikes or because of local dendritic processes. As a consequence, our model predicts that learning properties will be different at different locations on the dendritic tree. In conclusion, such site-specific synaptic plasticity would provide a neuron with powerful learning capabilities.

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49. Self-Organizing Dual Coding Based on Spike-Time-Dependent Plasticity

Naoki Masuda, Kazuyuki Aihara

https://doi.org/10.1162/089976604772744938 | pp. 627-663

It has been a matter of debate how firing rates or spatiotemporal spike patterns carry information in the brain. Recent experimental and theoretical work in part showed that these codes, especially a population rate code and a synchronous code, can be dually used in a single architecture. However, we are not yet able to relate the role of firing rates and synchrony to the spatiotemporal structure of inputs and the architecture of neural networks. In this article, we examine how feedforward neural networks encode multiple input

sources in the firing patterns. We apply spike-time-dependent plasticity as a fundamental mechanism to yield synaptic competition and the associated input filtering. We use the Fokker-Planck formalism to analyze the mechanism for synaptic competition in the case of multiple inputs, which underlies the formation of functional clusters in downstream layers in a self-organizing manner. Depending on the types of feedback coupling and shared connectivity, clusters are independently engaged in population rate coding or synchronous coding, or they interact to serve as input filters. Classes of dual codings and functional roles of spike-time-dependent plasticity are also discussed.

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50. The Shape of Neural Dependence

Rick L. Jenison, Richard A. Reale

https://doi.org/10.1162/089976604322860659 | pp. 665-672

The product-moment correlation coefficient is often viewed as a natural measure of dependence. However, this equivalence applies only in the context of elliptical distributions, most commonly the multivariate gaussian, where linear correlation indeed sufficiently describes the underlying dependence structure. Should the true probability distributions deviate from those with elliptical contours, linear correlation may convey misleading information on the actual underlying dependencies. It is often the case that probability distributions other than the gaussian distribution are necessary to properly capture the stochastic nature of single neurons, which as a consequence greatly complicates the construction of a flexible model of covariance. We show how arbitrary probability densities can be coupled to allow greater flexibility in the construction of multivariate neural population models.

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51. On the Phase Reduction and Response Dynamics of Neural Oscillator Populations

Eric Brown, Jeff Moehlis, Philip Holmes

https://doi.org/10.1162/089976604322860668 | pp. 673-715

We undertake a probabilistic analysis of the response of repetitively firing neural populations to simple pulselike stimuli. Recalling and extending results from the literature, we compute phase response curves (PRCs) valid near bifurcations to periodic firing for Hindmarsh-Rose, Hodgkin-Huxley, FitzHugh-Nagumo, and Morris-Lecar models, encompassing the four generic (codimension one) bifurcations. Phase density equations are then used to analyze the role of the bifurcation, and the resulting PRC, in responses to stimuli. In particular, we explore the interplay among stimulus duration, baseline firing frequency, and population-level response patterns. We interpret the results in terms of the signal processing measure of gain and discuss further applications and experimentally testable predictions.

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52. Estimating the Entropy Rate of Spike Trains via Lempel-Ziv Complexity

José M. Amigó, Janusz Szczepa■ski, Elek Wajnryb, María V. Sánchez-Vives

https://doi.org/10.1162/089976604322860677 | pp. 717-736

Normalized Lempel-Ziv complexity, which measures the generation rate of new patterns along a digital sequence, is closely related to such important source properties as entropy and compression ratio, but, in

contrast to these, it is a property of individual sequences. In this article, we propose to exploit this concept to estimate (or, at least, to bound from below) the entropy of neural discharges (spike trains). The main advantages of this method include fast convergence of the estimator (as supported by numerical simulation) and the fact that there is no need to know the probability law of the process generating the signal. Furthermore, we present numerical and experimental comparisons of the new method against the standard method based on word frequencies, providing evidence that this new approach is an alternative entropy estimator for binned spike trains.

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53. Investigation of Possible Neural Architectures Underlying Information-Geometric Measures

Masami Tatsuno, Masato Okada

https://doi.org/10.1162/089976604322860686 | pp. 737-765

A novel analytical method based on information geometry was recently proposed, and this method may provide useful insights into the statistical interactions within neural groups. The link between informationgeometric measures and the structure of neural interactions has not yet been elucidated, however, because of the ill-posed nature of the problem. Here, possible neural architectures underlying information-geometric measures are investigated using an isolated pair and an isolated triplet of model neurons. By assuming the existence of equilibrium states, we derive analytically the relationship between the information-geometric parameters and these simple neural architectures. For symmetric networks, the first- and second-order information-geometric parameters represent, respectively, the external input and the underlying connections between the neurons provided that the number of neurons used in the parameter estimation in the log-linear model and the number of neurons in the network are the same. For asymmetric networks, however, these parameters are dependent on both the intrinsic connections and the external inputs to each neuron. In addition, we derive the relation between the information-geometric parameter corresponding to the two-neuron interaction and a conventional cross-correlation measure. We also show that the information-geometric parameters vary depending on the number of neurons assumed for parameter estimation in the log-linear model. This finding suggests a need to examine the information-geometric method carefully. A possible criterion for choosing an appropriate orthogonal coordinate is also discussed. This article points out the importance of a model-based approach and sheds light on the possible neural structure underlying the application of information geometry to neural network analysis.

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54. Robustifying AdaBoost by Adding the Naive Error Rate

Takashi Takenouchi, Shinto Eguchi

https://doi.org/10.1162/089976604322860695 | pp. 767-787

AdaBoost can be derived by sequential minimization of the exponential loss function. It implements the learning process by exponentially reweighting examples according to classification results. However, weights are often too sharply tuned, so that AdaBoost suffers from the nonrobustness and overlearning. Wepropose a new boosting method that is a slight modification of AdaBoost. The loss function is defined by a mixture of the exponential loss and naive error loss functions. As a result, the proposed method incorporates the effect of forgetfulness into AdaBoost. The statistical significance of our method is

discussed, and simulations are presented for confirmation.

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55. Boosting with Noisy Data: Some Views from Statistical Theory

Wenxin Jiang

https://doi.org/10.1162/089976604322860703 | pp. 789-810

This letter is a comprehensive account of some recent findings about AdaBoost in the presence of noisy data when approached from the perspective of statistical theory. We start from the basic assumption of weak hypotheses used in AdaBoost and study its validity and implications on generalization error. We recommend studying the generalization error and comparing it to the optimal Bayes error when data are noisy. Analytic examples are provided to show that running the unmodified AdaBoost forever will lead to overfit. On the other hand, there exist regularized versions of AdaBoost that are consistent, in the sense that the resulting prediction will approximately attain the optimal performance in the limit of large training samples.

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56. Different Paradigms for Choosing Sequential Reweighting Algorithms

Gilles Blanchard

https://doi.org/10.1162/089976604322860712 | pp. 811-836

Analyses of the success of ensemble methods in classification have pointed out the important role played by the margin distribution function on the training and test sets. While it is acknowledged that one should generally try to achieve high margins on the training set, the more precise shape of the empirical margin distribution function one should favor in practice is subject to different approaches. We first present two concurrent philosophies for choosing the empirical margin profile: the minimax margin paradigm and the mean and variance paradigm. The best-known representative of the first paradigm is the AdaBoost algorithm, and this philosophy has been shown by several other authors to be closely related to the principle of the support vector machine. We show that the second paradigm is very close in spirit to Fisher's linear discriminant (in a feature space). We construct two boosting-type algorithms, very similar in their form, dedicated to one or the other philosophy. We consequently derive by interpolation a very simple family of iterative reweighting algorithms that can be understood as different trade-offs between the two paradigms and argue from experiments that this can allow for a suitable adaptivity to different classification problems, particularly in the presence of noise or excessive complexity of the base classifiers.

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57. Robust Formulations for Training Multilayer Perceptrons

Tommi Kärkkäinen, Erkki Heikkola

https://doi.org/10.1162/089976604322860721 | pp. 837-862

The connection between robust statistical estimates and nonsmooth optimization is established. Based on the resulting family of optimization problems, robust learning problem formulations with regularization-based control on the model complexity of the multilayer perceptron network are described and analyzed. Numerical experiments for simulated regression problems are conducted, and new strategies for determining the regularization coefficient are proposed and evaluated.

58. An Extended Projection Neural Network for Constrained Optimization

Youshen Xia

https://doi.org/10.1162/089976604322860730 | pp. 863-883

Recently, a projection neural network has been shown to be a promising computational model for solving variational inequality problems with box constraints. This letter presents an extended projection neural network for solving monotone variational inequality problems with linear and nonlinear constraints. In particular, the proposed neural network can include the projection neural network as a special case. Compared with the modified projection-type methods for solving constrained monotone variational inequality problems, the proposed neural network has a lower complexity and is suitable for parallel implementation. Furthermore, the proposed neural network is theoretically proven to be exponentially convergent to an exact solution without a Lipschitz condition. Illustrative examples show that the extended projection neural network can be used to solve constrained monotone variational inequality problems.

59. Spike-Timing-Dependent Plasticity: The Relationship to Rate-Based Learning for Models with Weight Dynamics Determined by a Stable Fixed Point

Anthony N. Burkitt, Hamish Meffin, David B. Grayden

https://doi.org/10.1162/089976604773135041 | pp. 885-940

Experimental evidence indicates that synaptic modification depends on the timing relationship between the presynaptic inputs and the output spikes that they generate. In this letter, results are presented for models of spike-timing-dependent plasticity (STDP) whose weight dynamics is determined by a stable fixed point. Four classes of STDP are identified on the basis of the time extent of their input-output interactions. The effect on the potentiation of synapses with different rates of input is investigated to elucidate the relationship of STDP with classical studies of long-term potentiation and depression and rate-based Hebbian learning. The selective potentiation of higher-rate synaptic inputs is found only for models where the time extent of the input-output interactions is input restricted (i.e., restricted to time domains delimited by adjacent synaptic inputs) and that have a time-asymmetric learning window with a longer time constant for depression than for potentiation. The analysis provides an account of learning dynamics determined by an input-selective stable fixed point. The effect of suppressive interspike interactions on STDP is also analyzed and shown to modify the synaptic dynamics.

60. Estimating the Temporal Interval Entropy of Neuronal Discharge

George N. Reeke, Allan D Coop

https://doi.org/10.1162/089976604773135050 | pp. 941-970

To better understand the role of timing in the function of the nervous system, we have developed a methodology that allows the entropy of neuronal discharge activity to be estimated from a spike train record when it may be assumed that successive interspike intervals are temporally uncorrelated. The so-called interval entropy obtained by this methodology is based on an implicit enumeration of all possible spike trains that are statistically indistinguishable from a given spike train. The interval entropy is calculated from an

analytic distribution whose parameters are obtained by maximum likelihood estimation from the interval probability distribution associated with a given spike train. We show that this approach reveals features of neuronal discharge not seen with two alternative methods of entropy estimation. The methodology allows for validation of the obtained data models by calculation of confidence intervals for the parameters of the analytic distribution and the testing of the significance of the fit between the observed and analytic interval distributions by means of Kolmogorov-Smirnov and Anderson-Darling statistics. The method is demonstrated by analysis of two different data sets: simulated spike trains evoked by either Poissonian or near-synchronous pulsed activation of a model cerebellar Purkinje neuron and spike trains obtained by extracellular recording from spontaneously discharging cultured rat hippocampal neurons.

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61. Dynamic Analysis of Neural Encoding by Point Process Adaptive Filtering

Uri T. Eden, Loren M. Frank, Riccardo Barbieri, Victor Solo, Emery N. Brown

https://doi.org/10.1162/089976604773135069 | pp. 971-998

Neural receptive fields are dynamic in that with experience, neurons change their spiking responses to relevant stimuli. To understand how neural systems adapt their representations of biological information, analyses of receptive field plasticity from experimental measurements are crucial. Adaptive signal processing, the well-established engineering discipline for characterizing the temporal evolution of system parameters, suggests a framework for studying the plasticity of receptive fields. We use the Bayes' rule Chapman-Kolmogorov paradigm with a linear state equation and point process observation models to derive adaptive filters appropriate for estimation from neural spike trains. We derive point process filter analogues of the Kalman filter, recursive least squares, and steepest-descent algorithms and describe the properties of these new filters. We illustrate our algorithms in two simulated data examples. The first is a study of slow and rapid evolution of spatial receptive fields in hippocampal neurons. The second is an adaptive decoding study in which a signal is decoded from ensemble neural spiking activity as the receptive fields of the neurons in the ensemble evolve. Our results provide a paradigm for adaptive estimation for point process observations and suggest a practical approach for constructing filtering algorithms to track neural receptive field dynamics on a millisecond timescale.

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62. Odor-Driven Attractor Dynamics in the Antennal Lobe Allow for Simple and Rapid Olfactory Pattern Classification

Roberto F. Galán, Silke Sachse, C. Giovanni Galizia, Andreas V. M. Herz

https://doi.org/10.1162/089976604773135078 | pp. 999-1012

The antennal lobe plays a central role for odor processing in insects, as demonstrated by electrophysiological and imaging experiments. Here we analyze the detailed temporal evolution of glomerular activity patterns in the antennal lobe of honeybees. We represent these spatiotemporal patterns as trajectories in a multidimensional space, where each dimension accounts for the activity of one glomerulus. Our data show that the trajectories reach odor-specific steady states (attractors) that correspond to stable activity patterns at about 1 second after stimulus onset. As revealed by a detailed mathematical investigation, the trajectories are characterized by different phases: response onset, steady-state plateau, response offset, and periods of spontaneous activity. An analysis based on support-vector machines quantifies the odor specificity of the attractors and the optimal time needed for odor

discrimination. The results support the hypothesis of a spatial olfactory code in the antennal lobe and suggest a perceptron-like readout mechanism that is biologically implemented in a downstream network, such as the mushroom body.

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63. Neural Mechanisms for the Robust Representation of Junctions

Thorsten Hansen, Heiko Neumann

https://doi.org/10.1162/089976604773135087 | pp. 1013-1037

Junctions provide important cues in various perceptual tasks, such as the determination of occlusion relationships for figure-ground separation, transparency perception, and object recognition, among others. In computer vision, junctions are used in a number of tasks, like point matching for image tracking or correspondence analysis. We propose a biologically motivated approach to junction representation in which junctions are implicitly characterized by high activity for multiple orientations within a cortical hypercolumn. A local measure of circular variance is suggested to extract junction points from this distributed representation. Initial orientation measurements are often fragmented and noisy. A coherent contour representation can be generated by a model of V1 utilizing mechanisms of collinear long-range integration and recurrent interaction. In the model, local oriented contrast estimates that are consistent within a more global context are enhanced while inconsistent activities are suppressed. In a series of computational experiments, we compare junction detection based on the new recurrent model with a feedforward model of complex cells. We show that localization accuracy and positive correctness in the detection of generic junction configurations such as L- and T-junctions is improved by the recurrent long-range interaction. Further, receiver operating characteristics analysis is used to evaluate the detection performance on both synthetic and camera images, showing the superior performance of the new approach. Overall, we propose that nonlocal interactions implemented by known mechanisms within V1 play an important role in detecting higher-order features such as corners and junctions.

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64. Greedy Learning of Multiple Objects in Images Using Robust Statistics and Factorial Learning

Christopher K. I. Williams, Michalis K. Titsias

https://doi.org/10.1162/089976604773135096 | pp. 1039-1062

We consider data that are images containing views of multiple objects. Our task is to learn about each of the objects present in the images. This task can be approached as a factorial learning problem, where each image must be explained by instantiating a model for each of the objects present with the correct instantiation parameters. A major problem with learning a factorial model is that as the number of objects increases, there is a combinatorial explosion of the number of configurations that need to be considered. We develop a method to extract object models sequentially from the data by making use of a robust statistical method, thus avoiding the combinatorial explosion, and present results showing successful extraction of objects from real images.

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65. Are Loss Functions All the Same?

Lorenzo Rosasco, Ernesto De Vito, Andrea Caponnetto, Michele Piana, Alessandro Verri
https://doi.org/10.1162/089976604773135104 | pp. 1063-1076

In this letter, we investigate the impact of choosing different loss functions from the viewpoint of statistical learning theory. We introduce a convexity assumption, which is met by all loss functions commonly used in the literature, and study how the bound on the estimation error changes with the loss. We also derive a general result on the minimizer of the expected risk for a convex loss function in the case of classification. The main outcome of our analysis is that for classification, the hinge loss appears to be the loss of choice. Other things being equal, the hinge loss leads to a convergence rate practically indistinguishable from the logistic loss rate and much better than the square loss rate. Furthermore, if the hypothesis space is sufficiently rich, the bounds obtained for the hinge loss are not loosened by the thresholding stage.

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66. Trading Variance Reduction with Unbiasedness: The Regularized Subspace Information Criterion for Robust Model Selection in Kernel Regression

Masashi Sugiyama, Motoaki Kawanabe, Klaus■Robert Müller
https://doi.org/10.1162/089976604773135113 | pp. 1077-1104

A well-known result by Stein (1956) shows that in particular situations, biased estimators can yield better parameter estimates than their generally preferred unbiased counterparts. This letter follows the same spirit, as we will stabilize the unbiased generalization error estimates by regularization and finally obtain more robust model selection criteria for learning. We trade a small bias against a larger variance reduction, which has the beneficial effect of being more precise on a single training set. We focus on the subspace information criterion (SIC), which is an unbiased estimator of the expected generalization error measured by the reproducing kernel Hilbert space norm. SIC can be applied to the kernel regression, and it was shown in earlier experiments that a small regularization of SIC has a stabilization effect. However, it remained open how to appropriately determine the degree of regularization in SIC. In this article, we derive an unbiased estimator of the expected squared error, between SIC and the expected generalization error and propose determining the degree of regularization of SIC such that the estimator of the expected squared error is minimized. Computer simulations with artificial and real data sets illustrate that the proposed method works effectively for improving the precision of SIC, especially in the high-noise-level cases. We furthermore compare the proposed method to the original SIC, the cross-validation, and an empirical Bayesian method in ridge parameter selection, with good results.

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67. Nonlinear Population Codes

Maoz Shamir, Haim Sompolinsky
https://doi.org/10.1162/089976604773717559 | pp. 1105-1136

Theoretical and experimental studies of distributed neuronal representations of sensory and behavioral variables usually assume that the tuning of the mean firing rates is the main source of information. However, recent theoretical studies have investigated the effect of cross-correlations in the trial-to-trial fluctuations of the neuronal responses on the accuracy of the representation. Assuming that only the first-order statistics of the neuronal responses are tuned to the stimulus, these studies have shown that in the presence of

correlations, similar to those observed experimentally in cortical ensembles of neurons, the amount of information in the population is limited, yielding nonzero error levels even in the limit of infinitely large populations of neurons. In this letter, we study correlated neuronal populations whose higher-order statistics, and in particular response variances, are also modulated by the stimulus. Weask two questions: Does the correlated noise limit the accuracy of the neuronal representation of the stimulus? and, How can a biological mechanism extract most of the information embedded in the higher-order statistics of the neuronal responses? Specifically, we address these questions in the context of a population of neurons coding an angular variable. We show that the information embedded in the variances grows linearly with the population size despite the presence of strong correlated noise. This information cannot be extracted by linear readout schemes, including the linear population vector. Instead, we propose a bilinear readout scheme that involves spatial decorrelation, quadratic nonlinearity, and population vector summation. We show that this nonlinear population vector scheme yields accurate estimates of stimulus parameters, with an efficiency that grows linearly with the population size. This code can be implemented using biologically plausible neurons.

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68. Enhancement of Information Transmission Efficiency by Synaptic Failures

Mark S. Goldman

https://doi.org/10.1162/089976604773717568 | pp. 1137-1162

Many synapses have a high percentage of synaptic transmission failures. I consider the hypothesis that synaptic failures can increase the efficiency of information transmission across the synapse. I use the information transmitted per vesicle release about the presynaptic spike train as a measure of synaptic transmission efficiency and show that this measure can increase with the synaptic failure probability. I analytically calculate the Shannon mutual information transmitted across two model synapses with probabilistic transmission: one with a constant probability of vesicle release and one with vesicle release probabilities governed by the dynamics of synaptic depression. For inputs generated by a non-Poisson process with positive autocorrelations, both synapses can transmit more information per vesicle release than a synapse with perfect transmission, although the information increases are greater for the depressing synapse than for a constant-probability synapse with the same average transmission probability. The enhanced performance of the depressing synapse over the constant-release-probability synapse primarily reflects a decrease in noise entropy rather than an increase in the total transmission entropy. This indicates a limitation of analysis methods, such as decorrelation, that consider only the total response entropy. My results suggest that synaptic transmission failures governed by appropriately tuned synaptic dynamics can increase the information-carrying efficiency of a synapse.

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69. An Algorithm for the Detection of Faces on the Basis of Gabor Features and Information Maximization

Hitoshi Imaoka, Kenji Okajima

https://doi.org/10.1162/089976604773717577 | pp. 1163-1191

We propose an algorithm for the detection of facial regions within input images. The characteristics of this algorithm are (1) a vast number of Gabor-type features (196,800) in various orientations, and with various frequencies and central positions, which are used as feature candidates in representing the patterns of an

image, and (2) an information maximization principle, which is used to select several hundred features that are suitable for the detection of faces from among these candidates. Using only the selected features in face detection leads to reduced computational cost and is also expected to reduce generalization error. We applied the system, after training, to 42 input images with complex backgrounds (Test Set A from the Carnegie Mellon University face data set). The result was a high detection rate of 87.0%, with only six false detections. We compared the result with other published face detection algorithms.

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70. Analysis of Sparse Representation and Blind Source Separation

Yuanqing Li, Andrzej Cichocki, ■■■-■■■ ■■■■■

https://doi.org/10.1162/089976604773717586 | pp. 1193-1234

In this letter, we analyze a two-stage cluster-then-l(1)-optimization approach for sparse representation of a data matrix, which is also a promising approach for blind source separation (BSS) in which fewer sensors than sources are present. First, sparse representation (factorization) of a data matrix is discussed. For a given overcomplete basis matrix, the corresponding sparse solution (coefficient matrix) with minimum l(1) norm is unique with probability one, which can be obtained using a standard linear programming algorithm. The equivalence of the l(1)-norm solution and the l(0)-norm solution is also analyzed according to a probabilistic framework. If the obtained l(1)-norm solution is sufficiently sparse, then it is equal to the l(0)-norm solution with a high probability. Furthermore, the l(1)- norm solution is robust to noise, but the l(0)-norm solution is not, showing that the l(1)-norm is a good sparsity measure. These results can be used as a recoverability analysis of BSS, as discussed. The basis matrix in this article is estimated using a clustering algorithm followed by normalization, in which the matrix columns are the cluster centers of normalized data column vectors. Zibulevsky, Pearlmutter, Boll, and Kisilev (2000) used this kind of two-stage approach in underdetermined BSS. Our recoverability analysis shows that this approach can deal with the situation in which the sources are overlapped to some degree in the analyzed domain and with the case in which the source number is unknown. It is also robust to additive noise and estimation error in the mixing matrix. Finally, four simulation examples and an EEG data analysis example are presented to illustrate the algorithm's utility and demonstrate its performance.

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71. Minimax Mutual Information Approach for Independent Component Analysis

Deniz Erdo■mu■, Kenneth E. Hild, Yadunandana N. Rao, José C. Pr■■ncipe

https://doi.org/10.1162/089976604773717595 | pp. 1235-1252

Minimum output mutual information is regarded as a natural criterion for independent component analysis (ICA) and is used as the performance measure in many ICA algorithms. Two common approaches in information-theoretic ICA algorithms are minimum mutual information and maximum output entropy approaches. In the former approach, we substitute some form of probability density function (pdf) estimate into the mutual information expression, and in the latter we incorporate the source pdf assumption in the algorithm through the use of nonlinearities matched to the corresponding cumulative density functions (cdf). Alternative solutions to ICA use higher-order cumulant-based optimization criteria, which are related to either one of these approaches through truncated series approximations for densities. In this article, we propose a new ICA algorithm motivated by the maximum entropy principle (for estimating signal

distributions). The optimality criterion is the minimum output mutual information, where the estimated pdfs are from the exponential family and are approximate solutions to a constrained entropy maximization problem. This approach yields an upper bound for the actual mutual information of the output signals - hence, the name minimax mutual information ICA algorithm. In addition, we demonstrate that for a specific selection of the constraint functions in the maximum entropy density estimation procedure, the algorithm relates strongly to ICA methods using higher-order cumulants.

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72. Improving Generalization Capabilities of Dynamic Neural Networks

M. Galicki, Lutz Leistritz, Ernst B. Zwick, Herbert Witte

https://doi.org/10.1162/089976604773717603 | pp. 1253-1282

This work addresses the problem of improving the generalization capabilities of continuous recurrent neural networks. The learning task is transformed into an optimal control framework in which the weights and the initial network state are treated as unknown controls. A new learning algorithm based on a variational formulation of Pontrayagin's maximum principle is proposed. Under reasonable assumptions, its convergence is discussed. Numerical examples are given that demonstrate an essential improvement of generalization capabilities after the learning process of a dynamic network.

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73. A Modified Algorithm for Generalized Discriminant Analysis

Wenming Zheng, Li Zhao, Cairong Zou

https://doi.org/10.1162/089976604773717612 | pp. 1283-1297

Generalized discriminant analysis (GDA) is an extension of the classical linear discriminant analysis (LDA) from linear domain to a nonlinear domain via the kernel trick. However, in the previous algorithm of GDA, the solutions may suffer from the degenerate eigenvalue problem (i.e., several eigenvectors with the same eigenvalue), which makes them not optimal in terms of the discriminant ability. In this letter, we propose a modified algorithm for GDA (MGDA) to solve this problem. The MGDA method aims to remove the degeneracy of GDA and find the optimal discriminant solutions, which maximize the between-class scatter in the subspace spanned by the degenerate eigenvectors of GDA. Theoretical analysis and experimental results on the ORL face database show that the MGDA method achieves better performance than the GDA method.

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74. Stability-Based Validation of Clustering Solutions

Tilman Lange, Volker Röth, Mikio L. Braun, Joachim M. Buhmann

https://doi.org/10.1162/089976604773717621 | pp. 1299-1323

Data clustering describes a set of frequently employed techniques in exploratory data analysis to extract "natural" group structure in data. Such groupings need to be validated to separate the signal in the data from spurious structure. In this context, finding an appropriate number of clusters is a particularly important model selection question. We introduce a measure of cluster stability to assess the validity of a cluster model. This stability measure quantifies the reproducibility of clustering solutions on a second sample, and it can be interpreted as a classification risk with regard to class labels produced by a clustering algorithm. The

preferred number of clusters is determined by minimizing this classification risk as a function of the number of clusters. Convincing results are achieved on simulated as well as gene expression data sets. Comparisons to other methods demonstrate the competitive performance of our method and its suitability as a general validation tool for clustering solutions in real-world problems.

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75. Bayesian Estimation of Stimulus Responses in Poisson Spike Trains

Sidney R. Lehky

https://doi.org/10.1162/089976604323057407 | pp. 1325-1343

A Bayesian method is developed for estimating neural responses to stimuli, using likelihood functions incorporating the assumption that spike trains follow either pure Poisson statistics or Poisson statistics with a refractory period. The Bayesian and standard estimates of the mean and variance of responses are similar and asymptotically converge as the size of the data sample increases. However, the Bayesian estimate of the variance of the variance is much lower. This allows the Bayesian method to provide more precise interval estimates of responses. Sensitivity of the Bayesian method to the Poisson assumption was tested by conducting simulations perturbing the Poisson spike trains with noise. This did not affect Bayesian estimates of mean and variance to a significant degree, indicating that the Bayesian method is robust. The Bayesian estimates were less affected by the presence of noise than estimates provided by the standard method.

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76. Comments on “A Parallel Mixture of SVMs for Very Large Scale Problems”

Xiaomei Liu, Lawrence Hall, Kevin W. Bowyer

https://doi.org/10.1162/089976604323057416 | pp. 1345-1351

Collobert, Bengio, and Bengio (2002) recently introduced a novel approach to using a neural network to provide a class prediction from an ensemble of support vector machines (SVMs). This approach has the advantage that the required computation scales well to very large data sets. Experiments on the Forest Cover data set show that this parallel mixture is more accurate than a single SVM, with 90.72% accuracy reported on an independent test set. Although this accuracy is impressive, their article does not consider alternative types of classifiers. We show that a simple ensemble of decision trees results in a higher accuracy, 94.75%, and is computationally efficient. This result is somewhat surprising and illustrates the general value of experimental comparisons using different types of classifiers.

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77. Automated Algorithms for Multiscale Morphometry of Neuronal Dendrites

Christina M. Weaver, Patrick R. Hof, Susan L. Wearne, W. Brent Lindquist

https://doi.org/10.1162/089976604323057425 | pp. 1353-1383

We describe the synthesis of automated neuron branching morphology and spine detection algorithms to provide multiscale three-dimensional morphological analysis of neurons. The resulting software is applied to the analysis of a high-resolution (0.098 microm x 0.098 microm x 0.081 microm) image of an entire pyramidal neuron from layer III of the superior temporal cortex in rhesus macaque monkey. The approach

provides a highly automated, complete morphological analysis of the entire neuron; each dendritic branch segment is characterized by several parameters, including branch order, length, and radius as a function of distance along the branch, as well as by the locations, lengths, shape classification (e.g., mushroom, stubby, thin), and density distribution of spines on the branch. Results for this automated analysis are compared to published results obtained by other computer-assisted manual means.

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78. Computing and Stability in Cortical Networks

Peter E. Latham, Sheila Nirenberg

https://doi.org/10.1162/089976604323057434 | pp. 1385-1412

Cortical neurons are predominantly excitatory and highly interconnected. In spite of this, the cortex is remarkably stable: normal brains do not exhibit the kind of runaway excitation one might expect of such a system. How does the cortex maintain stability in the face of this massive excitatory feedback? More importantly, how does it do so during computations, which necessarily involve elevated firing rates? Here we address these questions in the context of attractor networks-networks that exhibit multiple stable states, or memories. We find that such networks can be stabilized at the relatively low firing rates observed in vivo if two conditions are met: (1) the background state, where all neurons are firing at low rates, is inhibition dominated, and (2) the fraction of neurons involved in a memory is above some threshold, so that there is sufficient coupling between the memory neurons and the background. This allows "dynamical stabilization" of the attractors, meaning feedback from the pool of background neurons stabilizes what would otherwise be an unstable state. We suggest that dynamical stabilization may be a strategy used for a broad range of computations, not just those involving attractors.

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79. Real-Time Computation at the Edge of Chaos in Recurrent Neural Networks

Nils Bertschinger, Thomas Natschläger

https://doi.org/10.1162/089976604323057443 | pp. 1413-1436

Depending on the connectivity, recurrent networks of simple computational units can show very different types of dynamics, ranging from totally ordered to chaotic. We analyze how the type of dynamics (ordered or chaotic) exhibited by randomly connected networks of threshold gates driven by a time-varying input signal depends on the parameters describing the distribution of the connectivity matrix. In particular, we calculate the critical boundary in parameter space where the transition from ordered to chaotic dynamics takes place. Employing a recently developed framework for analyzing real-time computations, we show that only near the critical boundary can such networks perform complex computations on time series. Hence, this result strongly supports conjectures that dynamical systems that are capable of doing complex computational tasks should operate near the edge of chaos, that is, the transition from ordered to chaotic dynamics.

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80. Information Geometry of U-Boost and Bregman Divergence

Noboru Murata, Takashi Takenouchi, Takafumi Kanamori, Shinto Eguchi

https://doi.org/10.1162/089976604323057452 | pp. 1437-1481

We aim at an extension of AdaBoost to U-Boost, in the paradigm to build a stronger classification machine from a set of weak learning machines. A geometric understanding of the Bregman divergence defined by a generic convex function U leads to the U-Boost method in the framework of information geometry extended to the space of the finite measures over a label set. We propose two versions of U-Boost learning algorithms by taking account of whether the domain is restricted to the space of probability functions. In the sequential step, we observe that the two adjacent and the initial classifiers are associated with a right triangle in the scale via the Bregman divergence, called the Pythagorean relation. This leads to a mild convergence property of the U-Boost algorithm as seen in the expectation-maximization algorithm. Statistical discussions for consistency and robustness elucidate the properties of the U-Boost methods based on a stochastic assumption for training data.

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81. A New Approach to the Extraction of ANN Rules and to Their Generalization Capacity Through GP

Juan R. Rabuñal, Julián Dorado, Alejandro Pazos, Javier Pereira, Daniel Rivero

https://doi.org/10.1162/089976604323057461 | pp. 1483-1523

Various techniques for the extraction of ANN rules have been used, but most of them have focused on certain types of networks and their training. There are very few methods that deal with ANN rule extraction as systems that are independent of their architecture, training, and internal distribution of weights, connections, and activation functions. This article proposes a methodology for the extraction of ANN rules, regardless of their architecture, and based on genetic programming. The strategy is based on the previous algorithm and aims at achieving the generalization capacity that is characteristic of ANNs by means of symbolic rules that are understandable to human beings.

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82. A Classification Paradigm for Distributed Vertically Partitioned Data

Jayanta Kumar Basak, Ravi Kothari

https://doi.org/10.1162/089976604323057470 | pp. 1525-1544

In general, pattern classification algorithms assume that all the features are available during the construction of a classifier and its subsequent use. In many practical situations, data are recorded in different servers that are geographically apart, and each server observes features of local interest. The underlying infrastructure and other logistics (such as access control) in many cases do not permit continual synchronization. Each server thus has a partial view of the data in the sense that feature subsets (not necessarily disjoint) are available at each server. In this article, we present a classification algorithm for this distributed vertically partitioned data. We assume that local classifiers can be constructed based on the local partial views of the data available at each server. These local classifiers can be any one of the many standard classifiers (e.g., neural networks, decision tree, k nearest neighbor). Often these local classifiers are constructed to support decision making at each location, and our focus is not on these individual local classifiers. Rather, our focus is constructing a classifier that can use these local classifiers to achieve an error rate that is as close as possible to that of a classifier having access to the entire feature set. We empirically demonstrate the efficacy of the proposed algorithm and also provide theoretical results quantifying the loss that results as compared to the situation where the entire feature set is available to any single classifier.

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83. A Coarse-to-Fine Disparity Energy Model with Both Phase-Shift and Position-Shift Receptive Field Mechanisms

Yuzhi Chen, Ning Qian

https://doi.org/10.1162/089976604774201596 | pp. 1545-1577

Numerous studies suggest that the visual system uses both phase- and position-shift receptive field (RF) mechanisms for the processing of binocular disparity. Although the difference between these two mechanisms has been analyzed before, previous work mainly focused on disparity tuning curves instead of population responses. However, tuning curve and population response can exhibit different characteristics, and it is the latter that determines disparity estimation. Here we demonstrate, in the framework of the disparity energy model, that for relatively small disparities, the population response generated by the phase-shift mechanism is more reliable than that generated by the position-shift mechanism. This is true over a wide range of parameters, including the RF orientation. Since the phase model has its own drawbacks of underestimating large stimulus disparity and covering only a restricted range of disparity at a given scale, we propose a coarse-to-fine algorithm for disparity computation with a hybrid of phase-shift and position-shift components. In this algorithm, disparity at each scale is always estimated by the phase-shift mechanism to take advantage of its higher reliability. Since the phase-based estimation is most accurate at the smallest scale when the disparity is correspondingly small, the algorithm iteratively reduces the input disparity from coarse to fine scales by introducing a constant position-shift component to all cells for a given location in order to offset the stimulus disparity at that location. The model also incorporates orientation pooling and spatial pooling to further enhance reliability. We have tested the algorithm on both synthetic and natural stereo images and found that it often performs better than a simple scale-averaging procedure.

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84. A Preference for Phase-Based Disparity in a Neuromorphic Implementation of the Binocular Energy Model

E.K.C. Tsang, Bertram E. Shi

https://doi.org/10.1162/089976604774201604 | pp. 1579-1600

The relative depth of objects causes small shifts in the left and right retinal positions of these objects, called binocular disparity. This letter describes an electronic implementation of a single binocularly tuned complex cell based on the binocular energy model, which has been proposed to model disparity-tuned complex cells in the mammalian primary visual cortex. Our system consists of two silicon retinas representing the left and right eyes, two silicon chips containing retinotopic arrays of spiking neurons with monocular Gabor-type spatial receptive fields, and logic circuits that combine the spike outputs to compute a disparity-selective complex cell response. The tuned disparity can be adjusted electronically by introducing either position or phase shifts between the monocular receptive field profiles. Mismatch between the monocular receptive field profiles caused by transistor mismatch can degrade the relative responses of neurons tuned to different disparities. In our system, the relative responses between neurons tuned by phase encoding are better matched than neurons tuned by position encoding. Our numerical sensitivity analysis indicates that the relative responses of phase-encoded neurons that are least sensitive to the receptive field parameters vary the most in our system. We conjecture that this robustness may be one reason for the existence of phase-encoded disparity-tuned neurons in biological neural systems.

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85. Learning Classification in the Olfactory System of Insects

Ramón Huerta, Thomas Nowotny, Marta Garc■■a-Sánchez, Henry D. I. Abarbanel, M. I. Rabinovich

https://doi.org/10.1162/089976604774201613 | pp. 1601-1640

We propose a theoretical framework for odor classification in the olfactory system of insects. The classification task is accomplished in two steps. The first is a transformation from the antennal lobe to the intrinsic Kenyon cells in the mushroom body. This transformation into a higher-dimensional space is an injective function and can be implemented without any type of learning at the synaptic connections. In the second step, the encoded odors in the intrinsic Kenyon cells are linearly classified in the mushroom body lobes. The neurons that perform this linear classification are equivalent to hyperplanes whose connections are tuned by local Hebbian learning and by competition due to mutual inhibition. We calculate the range of values of activity and size of the network required to achieve efficient classification within this scheme in insect olfaction. We are able to demonstrate that biologically plausible control mechanisms can accomplish efficient classification of odors.

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86. Adaptive Blind Separation with an Unknown Number of Sources

Jimin Ye, Xiaolong Zhu, Xian■Da Zhang

https://doi.org/10.1162/089976604774201622 | pp. 1641-1660

The blind source separation (BSS) problem with an unknown number of sources is an important practical issue that is usually skipped by assuming that the source number n is known and equal to the number m of sensors. This letter studies the general BSS problem satisfying m ≥ n. First, it is shown that the mutual information of outputs of the separation network is a cost function for BSS, provided that the mixing matrix is of full column rank and the m×m separating matrix is nonsingular. The mutual information reaches its local minima at the separation points, where the m outputs consist of n desired source signals and m–n redundant signals. Second, it is proved that the natural gradient algorithm proposed primarily for complete BSS (m n) can be generalized to deal with the overdetermined BSS problem (m>n), but it would diverge inevitably due to lack of a stationary point. To overcome this shortcoming, we present a modified algorithm, which can perform BSS steadily and provide the desired source signals at specified channels if some matrix is designed properly. Finally, the validity of the proposed algorithm is confirmed by computer simulations on artificially synthesized data.

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87. Unsupervised Spike Detection and Sorting with Wavelets and Superparamagnetic Clustering

Rodrigo Quian Quiroga, Zoltán Nádasdy, Yoram Ben■Shaul

https://doi.org/10.1162/089976604774201631 | pp. 1661-1687

This study introduces a new method for detecting and sorting spikes from multiunit recordings. The method combines the wavelet transform, which localizes distinctive spike features, with superparamagnetic clustering, which allows automatic classification of the data without assumptions such as low variance or gaussian distributions. Moreover, an improved method for setting amplitude thresholds for spike detection is proposed. We describe several criteria for implementation that render the algorithm unsupervised and fast. The algorithm is compared to other conventional methods using several simulated data sets whose

characteristics closely resemble those of in vivo recordings. For these data sets, we found that the proposed algorithm outperformed conventional methods.

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88. Decomposition Methods for Linear Support Vector Machines

Wei■Chun Kao, Kai-Min Chung, Chia■Liang Sun, Chih■Jen Lin

https://doi.org/10.1162/089976604774201640 | pp. 1689-1704

In this letter, we show that decomposition methods with alpha seeding are extremely useful for solving a sequence of linear support vector machines (SVMs) with more data than attributes. This strategy is motivated by Keerthi and Lin (2003), who proved that for an SVM with data not linearly separable, after C is large enough, the dual solutions have the same free and bounded components. We explain why a direct use of decomposition methods for linear SVMs is sometimes very slow and then analyze why alpha seeding is much more effective for linear than nonlinear SVMs. We also conduct comparisons with other methods that are efficient for linear SVMs and demonstrate the effectiveness of alpha seeding techniques in model selection.

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89. An Asymptotic Statistical Theory of Polynomial Kernel Methods

Kazushi Ikeda

https://doi.org/10.1162/089976604774201659 | pp. 1705-1719

The generalization properties of learning classifiers with a polynomial kernel function are examined. In kernel methods, input vectors are mapped into a high-dimensional feature space where the mapped vectors are linearly separated. It is well-known that a linear dichotomy has an average generalization error or a learning curve proportional to the dimension of the input space and inversely proportional to the number of given examples in the asymptotic limit. However, it does not hold in the case of kernel methods since the feature vectors lie on a submanifold in the feature space, called the input surface. In this letter, we discuss how the asymptotic average generalization error depends on the relationship between the input surface and the true separating hyperplane in the feature space where the essential dimension of the true separating polynomial, named the class, is important. We show its upper bounds in several cases and confirm these using computer simulations.

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90. A Neural Root Finder of Polynomials Based on Root Moments

De-Shuang Huang, Horace H. S. Ip, Zheru Chi

https://doi.org/10.1162/089976604774201668 | pp. 1721-1762

This letter proposes a novel neural root finder based on the root moment method (RMM) to find the arbitrary roots (including complex ones) of arbitrary polynomials. This neural root finder (NRF) was designed based on feedforward neural networks (FNN) and trained with a constrained learning algorithm (CLA). Specifically, we have incorporated the a priori information about the root moments of polynomials into the conventional backpropagation algorithm (BPA), to construct a new CLA. The resulting NRF is shown to be able to rapidly estimate the distributions of roots of polynomials. We study and compare the advantage of the RMM-based NRF over the previous root coefficient method—based NRF and the traditional Muller and Laguerre

methods as well as the mathematica roots function, and the behaviors, the accuracies of the resulting root finders, and their training speeds of two specific structures corresponding to this FNN root finder: the log σand the σ FNN. We also analyze the effects of the three controlling parameters {δP 0 θ p η} with the CLA on the two NRFs theoretically and experimentally. Finally, we present computer simulation results to support our claims.

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91. A Note on the Applied Use of MDL Approximations

Danielle Navarro

https://doi.org/10.1162/0899766041336378 | pp. 1763-1768

An applied problem is discussed in which two nested psychological mod-els of retention are compared using minimum description length (MDL).The standard Fisher information approximation to the normalized maximum likelihood is calculated for these two models, with the result that the full model is assigned a smaller complexity, even for moderately large samples. A geometric interpretation for this behavior is considered, along with its practical implications.

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92. Optimal Reduced-Set Vectors for Support Vector Machines with a Quadratic Kernel

Thorsten Thies, F. Weber

https://doi.org/10.1162/0899766041336459 | pp. 1769-1777

To reduce computational cost, the discriminant function of a support vector machine (SVM) should be represented using as few vectors as possible. This problem has been tackled in different ways. In this article,we develop an explicit solution in the case of a general quadratic kernel k(x. x') = (C + D xT x')2. For a given number of vectors, this solution provides the best possible approximation and can even recover the discriminant function if the number of used vectors is large enough. The key idea is to express the inhomogeneous kernel as a homogeneous kernel ona space having one dimension more than the original one and to follow the approach of Burges (1996).

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93. Stochastic Reasoning, Free Energy, and Information Geometry

Shiro Ikeda, Toshiyuki Tanaka, ■■■-■■■ ■■■■■

https://doi.org/10.1162/0899766041336477 | pp. 1779-1810

Belief propagation (BP) is a universal method of stochastic reasoning. It gives exact inference for stochastic models with tree interactions and works surprisingly well even if the models have loopy interactions. Its performance has been analyzed separately in many fields, such as AI, statistical physics, information theory, and information geometry. This article gives a unified framework for understanding BP and related methods and summarizes the results obtained in many fields. In particular, BP and its variants, including tree reparameterization and concave-convex procedure, are reformulated with information-geometrical terms, and their relations to the free energy function are elucidated from an information-geometrical viewpoint. We then propose a family of new algorithms. The stabilities of the algorithms are analyzed, and methods to accelerate them are investigated.

94. Blind Separation of Positive Sources by Globally Convergent Gradient Search

Erkki Oja, Mark D. Plumbley

https://doi.org/10.1162/0899766041336413 | pp. 1811-1825

The instantaneous noise-free linear mixing model in independent component analysis is largely a solved problem under the usual assumption of independent nongaussian sources and full column rank mixing matrix. However, with some prior information on the sources, like positivity, new analysis and perhaps simplified solution methods may yet become possible. In this letter, we consider the task of independent component analysis when the independent sources are known to be nonnegative and well grounded, which means that they have a nonzero pdf in the region of zero. It can be shown that in this case, the solution method is basically very simple: an orthogonal rotation of the whitened observation vector into nonnegative outputs will give a positive permutation of the original sources. We propose a cost function whose minimum coincides with nonnegativity and derive the gradient algorithm under the whitening constraint, under which the separating matrix is orthogonal. We further prove that in the Stiefel manifold of orthogonal matrices, the cost function is a Lyapunov function for the matrix gradient flow, implying global convergence. Thus, this algorithm is guaranteed to find the nonnegative well-grounded independent sources. The analysis is complemented by a numerical simulation, which illustrates the algorithm.

95. A New Concept for Separability Problems in Blind Source Separation

Fabian J. Theis

https://doi.org/10.1162/0899766041336404 | pp. 1827-1850

The goal of blind source separation (BSS) lies in recovering the original independent sources of a mixed random vector without knowing the mixing structure. A key ingredient for performing BSS successfully is to know the indeterminacies of the problem-that is, to know how the separating model relates to the original mixing model (separability). For linear BSS, Comon (1994) showed using the Darmois-Skitovitch theorem that the linear mixing matrix can be found except for permutation and scaling. In this work, a much simpler, direct proof for linear separability is given. The idea is based on the fact that a random vector is independent if and only if the Hessian of its logarithmic density (resp. characteristic function) is diagonal everywhere. This property is then exploited to propose a new algorithm for performing BSS. Furthermore, first ideas of how to generalize separability results based on Hessian diagonalization to more complicated nonlinear models are studied in the setting of postnonlinear BSS.

96. Modeling Mental Navigation in Scenes with Multiple Objects

Patrick Byrne, Suzanna Becker

https://doi.org/10.1162/0899766041336468 | pp. 1851-1872

Various lines of evidence indicate that animals process spatial information regarding object locations differently from spatial information regarding environmental boundaries or landmarks. Following Wang and Spelke's (2002) observation that spatial updating of egocentric representations appears to lie at the heart of many navigational tasks in many species, including humans, we postulate a neural circuit that can support

this computation in parietal cortex, assuming that egocentric representations of multiple objects can be maintained in prefrontal cortex in spatial working memory (not simulated here). Our method is a generalization of an earlier model by Droulez and Berthoz (1991), with extensions to support observer rotation. We can thereby simulate perspective transformation of working memory representations of object coordinates based on an egomotion signal presumed to be generated via mental navigation. This biologically plausible transformation would allow a subject to recall the locations of previously viewed objects from novel viewpoints reached via imagined, discontinuous, or disoriented displacement. Finally, we discuss how this model can account for a wide range of experimental findings regarding memory for object locations, and we present several predictions made by the model.

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97. Failure of Motor Learning for Large Initial Errors

Terence D. Sanger

https://doi.org/10.1162/0899766041336431 | pp. 1873-1886

For certain complex motor tasks, humans may experience the frustration of a lack of improvement despite repeated practice. We investigate a computational basis for failure of motor learning when there is no prior information about the system to be controlled and when it is not practical to perform a thorough random exploration of the set of possible commands. In this case, if the desired movement has never yet been performed, then it may not be possible to learn the correct motor commands since there will be no appropriate training examples. We derive the mathematical basis for this phenomenon when the controller can be modeled as a linear combination of nonlinear basis functions trained using a gradient descent learning rule on the observed commands and their results. We show that there are two failure modes for which continued training examples will never lead to improvement in performance. We suggest that this may provide a model for the lack of improvement in human skills that can occur despite repeated practice of a complex task.

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98. Fair Attribution of Functional Contribution in Artificial and Biological Networks

Alon Keinan, Ben Sandbank, Claus C. Hilgetag, Isaac Meilijson, Eytan Ruppin

https://doi.org/10.1162/0899766041336387 | pp. 1887-1915

This letter presents the multi-perturbation Shapley value analysis (MSA), an axiomatic, scalable, and rigorous method for deducing causal function localization from multiple perturbations data. The MSA, based on fundamental concepts from game theory, accurately quantifies the contributions of network elements and their interactions, overcoming several shortcomings of previous function localization approaches. Its successful operation is demonstrated in both the analysis of a neurophysiological model and of reversible deactivation data. The MSA has a wide range of potential applications, including the analysis of reversible deactivation experiments, neuronal laser ablations, and transcranial magnetic stimulation "virtual lesions," as well as in providing insight into the inner workings of computational models of neurophysiological systems.

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99. Recurrent Network with Large Representational Capacity

Dražen Domijan

https://doi.org/10.1162/0899766041336422 | pp. 1917-1942

A recurrent network is proposed with the ability to bind image features into a unified surface representation within a single layer and without capacity limitations or border effects. A group of cells belonging to the same object or surface is labeled with the same activity amplitude, while cells in different groups are kept segregated due to lateral inhibition. Labeling is achieved by activity spreading through local excitatory connections. In order to prevent uncontrolled spreading, a separate network computes the intensity difference between neighboring locations and signals the presence of the surface boundary, which constrains local excitation. The quality of surface representation is not compromised due to the self-excitation. The model is also applied on gray-level images. In order to remove small, noisy regions, a feedforward network is proposed that computes the size of surfaces. Size estimation is based on the difference of dendritic inhibition in lateral excitatory and inhibitory pathways, which allows the network to selectively integrate signals only from cells with the same activity amplitude. When the output of the size estimation network is combined with the recurrent network, good segmentation results are obtained. Both networks are based on biophysically realistic mechanisms such as dendritic inhibition and multiplicative integration among different dendritic branches.

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100. A Solution for Two-Dimensional Mazes with Use of Chaotic Dynamics in a Recurrent Neural Network Model

Yoshikazu Suemitsu, Shigetoshi Nara

https://doi.org/10.1162/0899766041336440 | pp. 1943-1957

Chaotic dynamics introduced into a neural network model is applied to solving two-dimensional mazes, which are ill-posed problems. A moving object moves from the position at t to t + 1 by simply defined motion function calculated from firing patterns of the neural network model at each time step t. We have embedded several prototype attractors that correspond to the simple motion of the object orienting toward several directions in two-dimensional space in our neural network model. Introducing chaotic dynamics into the network gives outputs sampled from intermediate state points between embedded attractors in a state space, and these dynamics enable the object to move in various directions. System parameter switching between a chaotic and an attractor regime in the state space of the neural network enables the object to move to a set target in a two-dimensional maze. Results of computer simulations show that the success rate for this method over 300 trials is higher than that of random walk. To investigate why the proposed method gives better performance, we calculate and discuss statistical data with respect to dynamical structure.

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101. Online Adaptive Decision Trees

Jayanta Kumar Basak

https://doi.org/10.1162/0899766041336396 | pp. 1959-1981

Decision trees and neural networks are widely used tools for pattern classification. Decision trees provide highly localized representation, whereas neural networks provide a distributed but compact representation of the decision space. Decision trees cannot be induced in the online mode, and they are not adaptive to

changing environment, whereas neural networks are inherently capable of online learning and adpativity. Here we provide a classification scheme called online adaptive decision trees (OADT), which is a tree-structured network like the decision trees and capable of online learning like neural networks. A new objective measure is derived for supervised learning with OADT. Experimental results validate the effectiveness of the proposed classification scheme. Also, with certain real-life data sets, we find that OADT performs better than two widely used models: the hierarchical mixture of experts and multilayer perceptron.

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102. Understanding the Cortical Specialization for Horizontal Disparity

Jenny C. A. Read, Bruce G. Cumming

https://doi.org/10.1162/0899766041732440 | pp. 1983-2020

Because the eyes are displaced horizontally, binocular vision is inherently anisotropic. Recent experimental work has uncovered evidence of this anisotropy in primary visual cortex (V1): neurons respond over a wider range of horizontal than vertical disparity, regardless of their orientation tuning. This probably reflects the horizontally elongated distribution of two-dimensional disparity experienced by the visual system, but it conflicts with all existing models of disparity selectivity, in which the relative response range to vertical and horizontal disparities is determined by the preferred orientation. Potentially, this discrepancy could require us to abandon the widely held view that processing in V1 neurons is initially linear. Here, we show that these new experimental data can be reconciled with an initial linear stage; we present two physiologically plausible ways of extending existing models to achieve this. First, we allow neurons to receive input from multiple binocular subunits with different position disparities (previous models have assumed all subunits have identical position and phase disparity). Then we incorporate a form of divisive normalization, which has successfully explained many response properties of V1 neurons but has not previously been incorporated into a model of disparity selectivity. We show that either of these mechanisms decouples disparity tuning from orientation tuning and discuss how the models could be tested experimentally. This represents the first explanation of how the cortical specialization for horizontal disparity may be achieved.

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103. Different Predictions by the Minimum Variance and Minimum Torque-Change Models on the Skewness of Movement Velocity Profiles

Hirokazu Tanaka, Meihua Tai, Ning Qian

https://doi.org/10.1162/0899766041732431 | pp. 2021-2040

We investigated the differences between two well-known optimization principles for understanding movement planning: the minimum variance (MV) model of Harris and Wolpert (1998) and the minimum torque change (MTC) model of Uno, Kawato, and Suzuki (1989). Both models accurately describe the properties of human reaching movements in ordinary situations (e.g., nearly straight paths and bell-shaped velocity profiles). However, we found that the two models can make very different predictions when external forces are applied or when the movement duration is increased. We considered a second-order linear system for the motor plant that has been used previously to simulate eye movements and single-joint arm movements and were able to derive analytical solutions based on the MV and MTC assumptions. With the linear plant, the MTC model predicts that the movement velocity profile should always be symmetrical, independent of the external forces and movement duration. In contrast, the MV model strongly depends on the movement duration and the system's degree of stability; the latter in turn depends on the total forces.

The MV model thus predicts a skewed velocity profile under many circumstances. For example, it predicts that the peak location should be skewed toward the end of the movement when the movement duration is increased in the absence of any elastic force. It also predicts that with appropriate viscous and elastic forces applied to increase system stability, the velocity profile should be skewed toward the beginning of the movement. The velocity profiles predicted by the MV model can even show oscillations when the plant becomes highly oscillatory. Our analytical and simulation results suggest specific experiments for testing the validity of the two models.

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104. Disambiguating Visual Motion Through Contextual Feedback Modulation

Pierre Bayerl, Heiko Neumann

https://doi.org/10.1162/0899766041732404 | pp. 2041-2066

Motion of an extended boundary can be measured locally by neurons only orthogonal to its orientation (aperture problem) while this ambiguity is resolved for localized image features, such as corners or nonocclusion junctions. The integration of local motion signals sampled along the outline of a moving form reveals the object velocity. We propose a new model of V1-MT feedforward and feedback processing in which localized V1 motion signals are integrated along the feedforward path by model MT cells. Top-down feedback from MT cells in turn emphasizes model V1 motion activities of matching velocity by excitatory modulation and thus realizes an attentional gating mechanism. The model dynamics implement a guided filling-in process to disambiguate motion signals through biased on-center, off-surround competition. Our model makes predictions concerning the time course of cells in area MT and V1 and the disambiguation process of activity patterns in these areas and serves as a means to link physiological mechanisms with perceptual behavior. We further demonstrate that our model also successfully processes natural image sequences.

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105. Exact Solution for the Optimal Neuronal Layout Problem

Dmitri B. Chklovskii

https://doi.org/10.1162/0899766041732422 | pp. 2067-2078

Evolution perfected brain design by maximizing its functionality while minimizing costs associated with building and maintaining it. Assumption that brain functionality is specified by neuronal connectivity, implemented by costly biological wiring, leads to the following optimal design problem. For a given neuronal connectivity, find a spatial layout of neurons that minimizes the wiring cost. Unfortunately, this problem is difficult to solve because the number of possible layouts is often astronomically large. We argue that the wiring cost may scale as wire length squared, reducing the optimal layout problem to a constrained minimization of a quadratic form. For biologically plausible constraints, this problem has exact analytical solutions, which give reasonable approximations to actual layouts in the brain. These solutions make the inverse problem of inferring neuronal connectivity from neuronal layout more tractable.

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106. Decoding a Temporal Population Code

Philipp Knüsel, Reto Wyss, Peter König, Paul F. M. J. Verschure

https://doi.org/10.1162/0899766041732459 | pp. 2079-2100

Encoding of sensory events in internal states of the brain requires that this information can be decoded by other neural structures. The encoding of sensory events can involve both the spatial organization of neuronal activity and its temporal dynamics. Here we investigate the issue of decoding in the context of a recently proposed encoding scheme: the temporal population code. In this code, the geometric properties of visual stimuli become encoded into the temporal response characteristics of the summed activities of a population of cortical neurons. For its decoding, we evaluate a model based on the structure and dynamics of cortical microcircuits that is proposed for computations on continuous temporal streams: the liquid state machine. Employing the original proposal of the decoding network results in a moderate performance. Our analysis shows that the temporal mixing of subsequent stimuli results in a joint representation that compromises their classification. To overcome this problem, we investigate a number of initialization strategies. Whereas we observe that a deterministically initialized network results in the best performance, we find that in case the network is never reset, that is, it continuously processes the sequence of stimuli, the classification performance is greatly hampered by the mixing of information from past and present stimuli. We conclude that this problem of the mixing of temporally segregated information is not specific to this particular decoding model but relates to a general problem that any circuit that processes continuous streams of temporal information needs to solve. Furthermore, as both the encoding and decoding components of our network have been independently proposed as models of the cerebral cortex, our results suggest that the brain could solve the problem of temporal mixing by applying reset signals at stimulus onset, leading to a temporal segmentation of a continuous input stream.

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107. Minimal Models of Adapted Neuronal Response to In Vivo–Like Input Currents

Giancarlo La Camera, Alexander Rauch, Hans■R. Lüscher, Walter Senn, Stefano Fusi

https://doi.org/10.1162/0899766041732468 | pp. 2101-2124

Rate models are often used to study the behavior of large networks of spiking neurons. Here we propose a procedure to derive rate models that take into account the fluctuations of the input current and firing-rate adaptation, two ubiquitous features in the central nervous system that have been previously overlooked in constructing rate models. The procedure is general and applies to any model of firing unit. As examples, we apply it to the leaky integrate-and-fire (IF) neuron, the leaky IF neuron with reversal potentials, and to the quadratic IF neuron. Two mechanisms of adaptation are considered, one due to an afterhyperpolarization current and the other to an adapting threshold for spike emission. The parameters of these simple models can be tuned to match experimental data obtained from neocortical pyramidal neurons. Finally, we show how the stationary model can be used to predict the time-varying activity of a large population of adapting neurons.

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108. Including Long-Range Dependence in Integrate-and-Fire Models of the High Interspike-Interval Variability of Cortical Neurons

B. Scott Jackson

https://doi.org/10.1162/0899766041732413 | pp. 2125-2195

Many different types of integrate-and-fire models have been designed in order to explain how it is possible for a cortical neuron to integrate over many independent inputs while still producing highly variable spike trains. Within this context, the variability of spike trains has been almost exclusively measured using the coefficient of variation of interspike intervals. However, another important statistical property that has been found in cortical spike trains and is closely associated with their high firing variability is long-range dependence. We investigate the conditions, if any, under which such models produce output spike trains with both interspike-interval variability and long-range dependence similar to those that have previously been measured from actual cortical neurons. We first show analytically that a large class of high-variability integrate-and-fire models is incapable of producing such outputs based on the fact that their output spike trains are always mathematically equivalent to renewal processes. This class of models subsumes a majority of previously published models, including those that use excitation-inhibition balance, correlated inputs, partial reset, or nonlinear leakage to produce outputs with high variability. Next, we study integrate-and-fire models that have (nonPoissonian) renewal point process inputs instead of the Poisson point process inputs used in the preceding class of models. The confluence of our analytical and simulation results implies that the renewal-input model is capable of producing high variability and long-range dependence comparable to that seen in spike trains recorded from cortical neurons, but only if the interspike intervals of the inputs have infinite variance, a physiologically unrealistic condition. Finally, we suggest a new integrate-and-fire model that does not suffer any of the previously mentioned shortcomings. By analyzing simulation results for this model, we show that it is capable of producing output spike trains with interspike-interval variability and long-range dependence that match empirical data from cortical spike trains. This model is similar to the other models in this study, except that its inputs are fractional-gaussian-noise-driven Poisson processes rather than renewal point processes. In addition to this model's success in producing realistic output spike trains, its inputs have long-range dependence similar to that found in most subcortical neurons in sensory pathways, including the inputs to cortex. Analysis of output spike trains from simulations of this model also shows that a tight balance between the amounts of excitation and inhibition at the inputs to cortical neurons is not necessary for high interspike-interval variability at their outputs. Furthermore, in our analysis of this model, we show that the superposition of many fractional-gaussian-noise-driven Poisson processes does not approximate a Poisson process, which challenges the common assumption that the total effect of a large number of inputs on a neuron is well represented by a Poisson process.

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109. Learning Eigenfunctions Links Spectral Embedding and Kernel PCA

Yoshua Bengio, Olivier Delalleau, Nicolas Le Roux, Jean-François Paiement, Pascal Vincent, Marie Claude Ouimet

https://doi.org/10.1162/0899766041732396 | pp. 2197-2219

In this letter, we show a direct relation between spectral embedding methods and kernel principal components analysis and how both are special cases of a more general learning problem: learning the principal eigenfunctions of an operator defined from a kernel and the unknown data-generating density. Whereas spectral embedding methods provided only coordinates for the training points, the analysis justifies a simple extension to out-of-sample examples (the Nyström formula) for multidimensional scaling (MDS), spectral clustering, Laplacian eigenmaps, locally linear embedding (LLE), and Isomap. The analysis provides, for all such spectral embedding methods, the definition of a loss function, whose empirical average is minimized by the traditional algorithms. The asymptotic expected value of that loss defines a

generalization performance and clarifies what these algorithms are trying to learn. Experiments with LLE, Isomap, spectral clustering, and MDS show that this out-of-sample embedding formula generalizes well, with a level of error comparable to the effect of small perturbations of the training set on the embedding.

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110. Predicting Axonal Response to Molecular Gradients with a Computational Model of Filopodial Dynamics

Geoffrey J. Goodhill, Ming Gu, Jeffrey S. Urbach

https://doi.org/10.1162/0899766041941934 | pp. 2221-2243

Axons are often guided to their targets in the developing nervous system by attractive or repulsive molecular concentration gradients. We propose a computational model for gradient sensing and directed movement of the growth cone mediated by filopodia. We show that relatively simple mechanisms are sufficient to generate realistic trajectories for both the short-term response of axons to steep gradients and the long-term response of axons to shallow gradients. The model makes testable predictions for axonal response to attractive and repulsive gradients of different concentrations and steepness, the size of the intracellular amplification of the gradient signal, and the differences in intracellular signaling required for repulsive versus attractive turning.

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111. Insect-Inspired Estimation of Egomotion

Matthias Franz, Javaan Chahl, Holger G. Krapp

https://doi.org/10.1162/0899766041941899 | pp. 2245-2260

Tangential neurons in the fly brain are sensitive to the typical optic flow patterns generated during egomotion. In this study, we examine whether a simplified linear model based on the organization principles in tangential neurons can be used to estimate egomotion from the optic flow. We present a theory for the construction of an estimator consisting of a linear combination of optic flow vectors that incorporates prior knowledge about the distance distribution of the environment and about the noise and egomotion statistics of the sensor. The estimator is tested on a gantry carrying an omnidirectional vision sensor. The experiments show that the proposed approach leads to accurate and robust estimates of rotation rates, whereas translation estimates are of reasonable quality, albeit less reliable.

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112. Correlated Firing Improves Stimulus Discrimination in a Retinal Model

Garrett T. Kenyon, James Theiler, John S. George, B. J. Travis, David Marshak

https://doi.org/10.1162/0899766041941916 | pp. 2261-2291

Synchronous firing limits the amount of information that can be extracted by averaging the firing rates of similarly tuned neurons. Here, we show that the loss of such rate-coded information due to synchronous oscillations between retinal ganglion cells can be overcome by exploiting the information encoded by the correlations themselves. Two very different models, one based on axon-mediated inhibitory feedback and the other on oscillatory common input, were used to generate artificial spike trains whose synchronous oscillations were similar to those measured experimentally. Pooled spike trains were summed into a threshold detector whose output was classified using Bayesian discrimination. For a threshold detector with

short summation times, realistic oscillatory input yielded superior discrimination of stimulus intensity compared to rate-matched Poisson controls. Even for summation times too long to resolve synchronous inputs, gamma band oscillations still contributed to improved discrimination by reducing the total spike count variability, or Fano factor. In separate experiments in which neurons were synchronized in a stimulus-dependent manner without attendant oscillations, the Fano factor increased markedly with stimulus intensity, implying that stimulus-dependent oscillations can offset the increased variability due to synchrony alone.

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113. A Temporal Stability Approach to Position and Attention-Shift-Invariant Recognition

Muhua Li, James J. Clark

https://doi.org/10.1162/0899766041941907 | pp. 2293-2321

Incorporation of visual-related self-action signals can help neural networks learn invariance. We describe a method that can produce a network with invariance to changes in visual input caused by eye movements and covert attention shifts. Training of the network is controlled by signals associated with eye movements and covert attention shifting. A temporal perceptual stability constraint is used to drive the output of the network toward remaining constant across temporal sequences of saccadic motions and covert attention shifts. We use a four-layer neural network model to perform the position-invariant extraction of local features and temporal integration of invariant presentations of local features in a bottom-up structure. We present results on both simulated data and real images to demonstrate that our network can acquire both position and attention shift invariance.

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114. Testing for and Estimating Latency Effects for Poisson and Non-Poisson Spike Trains

Valérie Ventura

https://doi.org/10.1162/0899766041941952 | pp. 2323-2349

Determining the variations in response latency of one or several neurons to a stimulus is of interest in different contexts. Two common problems concern correlating latency with a particular behavior, for example, the reaction time to a stimulus, and adjusting tools for detecting synchronization between two neurons. We use two such problems to illustrate the latency testing and estimation methods developed in this article. Our test for latencies is a formal statistical test that produces a p-value. It is applicable for Poisson and non-Poisson spike trains via use of the bootstrap. Our estimation method is model free, it is fast and easy to implement, and its performance compares favorably to other methods currently available.

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115. Two-State Membrane Potential Fluctuations Driven by Weak Pairwise Correlations

Andrea Benucci, Paul F. M. J. Verschure, Peter König

https://doi.org/10.1162/0899766041941871 | pp. 2351-2378

Physiological experiments demonstrate the existence of weak pairwise correlations of neuronal activity in mammalian cortex (Singer, 1993). The functional implications of this correlated activity are hotly debated (Roskies et al., 1999). Nevertheless, it is generally considered a widespread feature of cortical dynamics. In recent years, another line of research has attracted great interest: the observation of a bimodal distribution of the membrane potential defining up states and down states at the single cell level (Wilson & Kawaguchi, 1996; Steriade, Contreras, & Amzica, 1994; Contreras & Steriade, 1995; Steriade, 2001). Here we use a theoretical approach to demonstrate that the latter phenomenon is a natural consequence of the former. In particular, we show that weak pairwise correlations of the inputs to a compartmental model of a layer V pyramidal cell can induce bimodality in its membrane potential. We show how this relationship can account for the observed increase of the power in the gamma-frequency band during up states, as well as the increase in the standard deviation and fraction of time spent in the depolarized state (Anderson, Lampl, Reichova, Carandini, & Ferster, 2000). In order to quantify the relationship between the correlation properties of a cortical network and the bistable dynamics of single neurons, we introduce a number of new indices. Subsequently, we demonstrate that a quantitative agreement with the experimental data can be achieved, introducing voltage-dependent mechanisms in our neuronal model such as Ca(2+)- and Ca(2+)-dependent K(+) channels. In addition, we show that the up states and down states of the membrane potential are dependent on the dendritic morphology of cortical neurons. Furthermore, bringing together network and single cell dynamics under a unified view allows the direct transfer of results obtained in one context to the other and suggests a new experimental paradigm: the use of specific intracellular analysis as a powerful tool to reveal the properties of the correlation structure present in the network dynamics.

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116. On the Uniqueness of Loopy Belief Propagation Fixed Points

Tom Heskes

https://doi.org/10.1162/0899766041941943 | pp. 2379-2413

We derive sufficient conditions for the uniqueness of loopy belief propagation fixed points. These conditions depend on both the structure of the graph and the strength of the potentials and naturally extend those for convexity of the Bethe free energy. We compare them with (a strengthened version of) conditions derived elsewhere for pairwise potentials. We discuss possible implications for convergent algorithms, as well as for other approximate free energies.

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117. Neural Network Uncertainty Assessment Using Bayesian Statistics: A Remote Sensing Application

Filipe Aires, Catherine Prigent, William B. Rossow

https://doi.org/10.1162/0899766041941925 | pp. 2415-2458

Neural network (NN) techniques have proved successful for many regression problems, in particular for remote sensing; however, uncertainty estimates are rarely provided. In this article, a Bayesian technique to evaluate uncertainties of the NN parameters (i.e., synaptic weights) is first presented. In contrast to more traditional approaches based on point estimation of the NN weights, we assess uncertainties on such estimates to monitor the robustness of the NN model. These theoretical developments are illustrated by applying them to the problem of retrieving surface skin temperature, microwave surface emissivities, and integrated water vapor content from a combined analysis of satellite microwave and infrared observations

over land. The weight uncertainty estimates are then used to compute analytically the uncertainties in the network outputs (i.e., error bars and correlation structure of these errors). Such quantities are very important for evaluating any application of an NN model. The uncertainties on the NN Jacobians are then considered in the third part of this article. Used for regression fitting, NN models can be used effectively to represent highly nonlinear, multivariate functions. In this situation, most emphasis is put on estimating the output errors, but almost no attention has been given to errors associated with the internal structure of the regression model. The complex structure of dependency inside the NN is the essence of the model, and assessing its quality, coherency, and physical character makes all the difference between a blackbox model with small output errors and a reliable, robust, and physically coherent model. Such dependency structures are described to the first order by the NN Jacobians: they indicate the sensitivity of one output with respect to the inputs of the model for given input data. We use a Monte Carlo integration procedure to estimate the robustness of the NN Jacobians. A regularization strategy based on principal component analysis is proposed to suppress the multicollinearities in order to make these Jacobians robust and physically meaningful.

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118. Principal Components Analysis Competitive Learning

Ezequiel López■Rubio, Juan Miguel Ortiz-de-Lazcano-Lobato, José Muñoz Pérez, José Antonio Gómez■Ruiz

https://doi.org/10.1162/0899766041941880 | pp. 2459-2481

We present a new neural model that extends the classical competitive learning by performing a principal components analysis (PCA) at each neuron. This model represents an improvement with respect to known local PCA methods, because it is not needed to present the entire data set to the network on each computing step. This allows a fast execution while retaining the dimensionality-reduction properties of the PCA. Furthermore, every neuron is able to modify its behavior to adapt to the local dimensionality of the input distribution. Hence, our model has a dimensionality estimation capability. The experimental results we present show the dimensionality-reduction capabilities of the model with multisensor images.

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119. How Many Clusters? An Information-Theoretic Perspective

Susanne Still, William Bialek

https://doi.org/10.1162/0899766042321751 | pp. 2483-2506

Clustering provides a common means of identifying structure in complex data, and there is renewed interest in clustering as a tool for the analysis of large data sets in many fields. A natural question is how many clusters are appropriate for the description of a given system. Traditional approaches to this problem are based on either a framework in which clusters of a particular shape are assumed as a model of the system or on a two-step procedure in which a clustering criterion determines the optimal assignments for a given number of clusters and a separate criterion measures the goodness of the classification to determine the number of clusters. In a statistical mechanics approach, clustering can be seen as a trade-off between energy- and entropy-like terms, with lower temperature driving the proliferation of clusters to provide a more detailed description of the data. For finite data sets, we expect that there is a limit to the meaningful structure that can be resolved and therefore a minimum temperature beyond which we will capture sampling noise. This suggests that correcting the clustering criterion for the bias that arises due to sampling errors will allow

us to find a clustering solution at a temperature that is optimal in the sense that we capture maximal meaningful structure--without having to define an external criterion for the goodness or stability of the clustering. We show that in a general information-theoretic framework, the finite size of a data set determines an optimal temperature, and we introduce a method for finding the maximal number of clusters that can be resolved from the data in the hard clustering limit.

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120. A Learning Rule for Local Synaptic Interactions Between Excitation and Shunting Inhibition

Chunhui Mo, Ming Gu, Christof Koch

https://doi.org/10.1162/0899766042321788 | pp. 2507-2532

The basic requirement for direction selectivity is a nonlinear interaction between two different inputs in space-time. In some models, the interaction is hypothesized to occur between excitation and inhibition of the shunting type in the neuron's dendritic tree. How can the required spatial specificity be acquired in an unsupervised manner? We here propose an activity-based, local learning model that can account for direction selectivity in visual cortex based on such a local veto operation and that depends on synaptically induced changes in intracellular calcium concentration. Our biophysical simulations suggest that a model cell with our learning algorithm can develop direction selectivity organically after unsupervised training. The learning rule is also applicable to a neuron with multiple-direction-selective subunits and to a pair of cells with opposite-direction selectivities and is stable under different starting conditions, delays, and velocities.

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121. Maximum Likelihood Estimation of a Stochastic Integrate-and-Fire Neural Encoding Model

Liam Paninski, Jonathan W. Pillow, Eero P. Simoncelli

https://doi.org/10.1162/0899766042321797 | pp. 2533-2561

We examine a cascade encoding model for neural response in which a linear filtering stage is followed by a noisy, leaky, integrate-and-fire spike generation mechanism. This model provides a biophysically more realistic alternative to models based on Poisson (memoryless) spike generation, and can effectively reproduce a variety of spiking behaviors seen in vivo. We describe the maximum likelihood estimator for the model parameters, given only extracellular spike train responses (not intracellular voltage data). Specifically, we prove that the log-likelihood function is concave and thus has an essentially unique global maximum that can be found using gradient ascent techniques. We develop an efficient algorithm for computing the maximum likelihood solution, demonstrate the effectiveness of the resulting estimator with numerical simulations, and discuss a method of testing the model's validity using time-rescaling and density evolution techniques.

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122. Image Representation by Complex Cell Responses

Ingo J. Wundrich, Christoph von der Malsburg, Rolf P. Würtz

https://doi.org/10.1162/0899766042321760 | pp. 2563-2575

We present an analysis of the representation of images as the magnitudes of their transform with complex-valued Gabor wavelets. Such a representation is a model for complex cells in the early stage of visual processing and of high technical usefulness for image understanding, because it makes the representation insensitive to small local shifts. We show that if the images are band limited and of zero mean, then reconstruction from the magnitudes is unique up to the sign for almost all images.

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123. Modeling of Synchronized Bursting Events: The Importance of Inhomogeneity

Erez Persi, D. Horn, Vladislav Volman, Ronen Segev, Eshel Ben■Jacob

https://doi.org/10.1162/0899766042321823 | pp. 2577-2595

Cultured in vitro neuronal networks are known to exhibit synchronized bursting events (SBE), during which most of the neurons in the system spike within a time window of approximately 100 msec. Such phenomena can be obtained in model networks based on Markram-Tsodyks frequency-dependent synapses. In order to account correctly for the detailed behavior of SBEs, several modifications have to be implemented in such models. Random input currents have to be introduced to account for the rising profile of SBEs. Dynamic thresholds and inhomogeneity in the distribution of neuronal resistances enable us to describe the profile of activity within the SBE and the heavy-tailed distributions of interspike intervals and interevent intervals. Thus, we can account for the interesting appearance of Levy distributions in the data.

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124. Mean Field and Capacity in Realistic Networks of Spiking Neurons Storing Sparsely Coded Random Memories

Emanuele Curti, Gianluigi Mongillo, Giancarlo La Camera, Daniel J. Amit

https://doi.org/10.1162/0899766042321805 | pp. 2597-2637

Mean-field (MF) theory is extended to realistic networks of spiking neurons storing in synaptic couplings of randomly chosen stimuli of a given low coding level. The underlying synaptic matrix is the result of a generic, slow, long-term synaptic plasticity of two-state synapses, upon repeated presentation of the fixed set of the stimuli to be stored. The neural populations subtending the MF description are classified by the number of stimuli to which their neurons are responsive (multiplicity). This involves 2p + 1 populations for a network storing p memories. The computational complexity of the MF description is then significantly reduced by observing that at low coding levels (f), only a few populations remain relevant: the population of mean multiplicity - pf and those of multiplicity of order square root pf around the mean. The theory is used to produce (predict) bifurcation diagrams (the onset of selective delay activity and the rates in its various stationary states) and to compute the storage capacity of the network (the maximal number of single items used in training for each of which the network can sustain a persistent, selective activity state). This is done in various regions of the space of constitutive parameters for the neurons and for the learning process. The capacity is computed in MF versus potentiation amplitude, ratio of potentiation to depression probability and coding level f. The MF results compare well with recordings of delay activity rate distributions in simulations of the underlying microscopic network of 10,000 neurons.

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125. Canonical Correlation Analysis: An Overview with Application to Learning Methods

David R. Hardoon, Sándor Szedmák, John Shawe■Taylor

https://doi.org/10.1162/0899766042321814 | pp. 2639-2664

We present a general method using kernel canonical correlation analysis to learn a semantic representation to web images and their associated text. The semantic space provides a common representation and enables a comparison between the text and images. In the experiments, we look at two approaches of retrieving images based on only their content from a text query. We compare orthogonalization approaches against a standard cross-representation retrieval technique known as the generalized vector space model.

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126. Cognitive Map Formation Through Sequence Encoding by Theta Phase Precession

Hiroaki Wagatsuma, Yoko Yamaguchi

https://doi.org/10.1162/0899766042321742 | pp. 2665-2697

The rodent hippocampus has been thought to represent the spatial environment as a cognitive map. The associative connections in the hippocampus imply that a neural entity represents the map as a geometrical network of hippocampal cells in terms of a chart. According to recent experimental observations, the cells fire successively relative to the theta oscillation of the local field potential, called theta phase precession, when the animal is running. This observation suggests the learning of temporal sequences with asymmetric connections in the hippocampus, but it also gives rather inconsistent implications on the formation of the chart that should consist of symmetric connections for space coding. In this study, we hypothesize that the chart is generated with theta phase coding through the integration of asymmetric connections. Our computer experiments use a hippocampal network model to demonstrate that a geometrical network is formed through running experiences in a few minutes. Asymmetric connections are found to remain and distribute heterogeneously in the network. The obtained network exhibits the spatial localization of activities at each instance as the chart does and their propagation that represents behavioral motions with multidirectional properties. We conclude that theta phase precession and the Hebbian rule with a time delay can provide the neural principles for learning the cognitive map.

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127. A Complex-Valued RTRL Algorithm for Recurrent Neural Networks

Su Lee Goh, Danilo P. Mandic

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A complex-valued real-time recurrent learning (CRTRL) algorithm for the class of nonlinear adaptive filters realized as fully connected recurrent neural networks is introduced. The proposed CRTRL is derived for a general complex activation function of a neuron, which makes it suitable for nonlinear adaptive filtering of complex-valued nonlinear and nonstationary signals and complex signals with strong component correlations. In addition, this algorithm is generic and represents a natural extension of the real-valued RTRL. Simulations on benchmark and real-world complex-valued signals support the approach.

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